

EDUCATIVE COMMENTARY ON ISI 2015 MATHEMATICS PAPERS

(Last Updated on 22/04/2018)

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Introduction

This is a **revised** version of the commentary of ISI 2015 entrance test for B.Sc. The last version was uploaded on 23/08/2017. Apart from correcting a few minor errors, the only significant change in the present version is that after giving the solution to Q.8 of Paper 2 (the best question in the entire paper), an alternative short solution is given to the first part which makes a crucial use of the data that the terms of the given progression are reciprocals of integers.

Readers are invited to point out errors, if any, in Paper 1 as well as Paper 2 commentary.

By and large Paper 1 problems were disappointing. There was really not much to write about most of them. So I have filled the gap by writing short essays on some peripherally related topics like orthogonal trajectories (Q.7) continuous growth (Q.8) and Fermat's two square theorem (Q.12).

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, published by Universities Press, Hyderabad.

ISI BStat-BMath-UGA-2015 Paper with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Let $\mathbb{C}$ denote the set of complex numbers and $S = \{z \in \mathbb{C} : \bar{z} = z^2\}$ where $\bar{z}$ denote the complex conjugate of z . Then S has

- | | |
|-------------------|--------------------|
| (A) two elements | (B) three elements |
| (C) four elements | (D) six elements. |

Answer and Comments: (C). Obviously, $0 \in S$. To find the non-zero elements in S take the absolute values of both the sides of the given equation to get $|z| = |z|^2$ and hence $|z| = 1$. This implies $\bar{z} = \frac{1}{z}$ and the given equation becomes $z^3 = 1$, which has three solutions $1, \omega$ and ω^2 . So, in all S has four elements. ■

An absolutely simple problem. Those who know about ω , the complex cube root of unity, will recognise it immediately as one of the solutions of the equation $\bar{z} = z^2$ and that will lead to the answer more easily. Those who miss 0, will pay a zero price because there is no provision for negative marking, unlike in JEE!

Q.2 The number of one-to-one functions from a set with 3 elements to a set with 6 elements is

- (A) 20 (B) 120 (C) 216 (D) 720

Answer and Comments: (B). $6 \times 5 \times 4 = 120$. ■

The only comment that comes to mind is that it is hard to believe that a test whose counterparts in the immediately preceding and following years (viz. 2014 and 2016 respectively) carried many thought provoking questions would contain such a question which is such a direct application of one of the very first rules in combinatorics, viz. the product rule. Even JEE can't beat this!

Q.3 Two sides of a triangle are of length 2 cm. and 3 cm. Then the maximum possible area (in cm^2) of the triangle is:

- (A) 2 (B) 3 (C) 4 (D) 6

Answer and Comments: (B). Yet another single rule based problem. If the angle between the given sides is θ , then its area is $\frac{1}{2} \times 2 \times 3 \times \sin \theta = 3 \sin \theta$ whose maximum value is 3, occurring when $\theta = 90^\circ$. ■

In terms of degree of thinking and computations needed, this problem is antithetical to Q.27 in 2014 ISI which asked to find the minimum value of $|\sin x + \cos x + \tan x + \cot x + \sec x + \csc x|$. It is possible that nobody came up with the correct answer to it (viz. $2\sqrt{2} - 1$) and this might have kindled some rethinking about the efficacy of asking such gems in an MCQ test. If a problem is so difficult that hardly anybody can do it, it enhances the prestige of the test in academic circles, since an outsider gets the impression that you have to be exceptionally bright to crack it. But it makes little dent on the actual selection which is by a competitive process. Moderately difficult problems which, say, 20% of the candidates can solve are ideal for such selections.

While such a rethinking (if at all it was done) is a commendable step, it hardly meant that the paper setters should go to the opposite extreme by asking a problem which 90% of the candidates can solve comfortably!

Q.4 The number of factors of $2^{15} \times 3^{10} \times 5^6$ which are either perfect squares or perfect cubes (or both) is

- (A) 252 (B) 265 (C) 260 (D) 264

Answer and Comments: (A). Let F be the set of all possible (positive) factors of the given number. Each such factor is of the form $2^r 3^s 5^t$ where r, s, t are integers satisfying

$$0 \leq r \leq 15 \quad 0 \leq s \leq 10 \quad \text{and} \quad 0 \leq t \leq 6 \quad (1)$$

So every factor can be identified with a unique ordered triple (r, s, t) satisfying the conditions in (1). Although not asked in the problem, we

get that $|F|$ (i.e. the number of elements in the set F) as

$$|F| = 16 \times 11 \times 7 = 1232 \quad (2)$$

Now let A be the subset of F consisting of those factors which are perfect squares and B be the subset consisting of those factors that are perfect cubes. Then the question asks for $|A \cup B|$. By a very standard formula, called the **principle of inclusion and exclusion**, or, simply by common sense,

$$|A \cup B| = |A| + |B| - |A \cap B| \quad (3)$$

This reduces the problem to counting each of the terms on the R.H.S. A factor $2^r 3^s 5^t$ is a perfect square if and only if r, s, t are all even. in view of (1), r can take the values 0, 2, 4, 6, 8, 10, 12 and 14, i.e. 8 possible values in all. Similarly s can 6 possible values and t can take 3 possible values. Hence by the product rule (which was the sole idea in Q.2), we get

$$|A| = 8 \times 6 \times 4 = 192 \quad (4)$$

Similarly, for $2^r 3^s 5^t$ to be a perfect cube, each r, s, t must be a multiple of 3 subject to (1). That gives

$$|B| = 6 \times 4 \times 3 = 72 \quad (5)$$

Finally, for the factor $2^r 3^s 5^t$ to be both a square and a cube, each of r, s, t is a multiple of 6. So,

$$|A \cap B| = 3 \times 2 \times 2 = 12 \quad (6)$$

Straight substitution into (3) gives

$$|A \cup B| = 192 + 72 - 12 = 252 \quad (7)$$

which completes the solution.■

By giving the prime power factorisation of the number, the paper-setters have spared the candidates a lot of drudgery. What remains is an application of the principle of inclusion and exclusion. There is some duplication of work in getting the three counts and that increases the chances of numerical errors. Still, it is a reasonable question. Mathematically, this question is far more involved than Q.2. There was no point in asking Q.2 in presence of the present question.

Q.5. The minimum value of the function $f(x) = x^2 + 4x + \frac{4}{x} + \frac{1}{x^2}$ where $x > 0$ is :

- (A) 9.5 (B) 10 (C) 15 (D) 20

Answer and Comments: (B). The last question was a welcome change from the pattern of single idea based questions. With this question, we are back into it. The only difference is that this time the single idea needed is the A.M.-G.M. inequality. Applying it, the sum $x^2 + \frac{1}{x^2}$ is minimum when $x = 1$. Similarly $x + \frac{1}{x}$ is minimum when $x = 1$. As both these minima occur at the same value of x , the minimum of the expression also occurs at $x = 1$ and the minimum value is 10. ■

Problems based on the A.M.-G.M. inequality are very common in JEE and hence in most its sister examinations. Having lost its novelty, hardly a question is asked where it is applied in such an obvious manner. If one is asked to minimise $2 \sec x + \cos^2 x$, the A.M.-G.M. inequality for two terms does not help. It is only when you write it as a sum of three terms as $\sec x + \sec x + \cos^2 x$ that the minimum is 3 because the G.M. of the three numbers $\sec x, \sec x$ and $\cos^2 x$ is 1. Yet another disappointing problem.

Q.6 The minimum area of the triangle formed by any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the coordinate axes is

- (A) ab (B) $\frac{a^2+b^2}{2}$ (C) $\frac{(a+b)^2}{4}$ (D) $\frac{a^2+ab+b^2}{3}$

Answer and Comments: (A). A typical point P on the ellipse is of the form

$$x = a \cos \theta, \quad y = b \sin \theta \quad (1)$$

for some $\theta \in [0, 2\pi]$. By symmetry, we may suppose P lies in the first quadrant, i.e. $\theta \in [0, \pi/2]$. The slope of the tangent at this point is $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = -\frac{b}{a} \cot \theta$. So the equation of the tangent at P is

$$y - b \sin \theta = -\frac{b}{a} \cot \theta (x - a \cos \theta) \quad (2)$$

Its x -intercept, say A , is obtained by setting $y = 0$. This gives

$$A = a(\cos \theta + \sin \theta \tan \theta) = a \sec \theta \quad (3)$$

Similarly, the y -intercept, say B is

$$B = b(\sin \theta + \cos \theta \cot \theta) = b \operatorname{cosec} \theta \quad (4)$$

One can also get these from the readymade formula for the equation of the tangent at a point in the parametric form, viz.

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad (5)$$

The area, say Δ of the triangle described in the question is $\frac{1}{2}|A||B|$. But as we are assuming that $\theta \in [0, \pi/2]$, both A, B are positive. So,

$$\Delta = \frac{1}{2}AB = \frac{ab}{2} \sec \theta \operatorname{cosec} \theta \quad (6)$$

We have to minimise this as a function of θ . (Note that there is no maximum, because as P nears the end of either of the axes, the tangent at P is nearly parallel to one of the axis and so the triangle has a very large area.)

We can use calculus to minimise Δ . But a simpler way is to write

$$\Delta = \frac{ab}{2 \sin \theta \cos \theta} = \frac{ab}{\sin 2\theta} \quad (7)$$

The minimum occurs when $\sin 2\theta$ is maximum, i.e. 1. So the minimum value of Δ is ab . ■

A straightforward problem. The computations are reasonable. A good candidate can save some time by writing down (5) directly. But the saving is not substantial.

Q.7 The angle between the hyperbolas $xy = 1$ and $x^2 - y^2 = 1$ (at their point of intersection) is

$$(A) \frac{\pi}{2} \quad (B) \frac{\pi}{3} \quad (C) \frac{\pi}{4} \quad (D) \frac{\pi}{6}$$

Answer and Comments: (A). To find the points of intersection, we need to solve the following system of simultaneous equations

$$x^2 - y^2 = 1 \quad (1)$$

$$\text{and } xy = 1 \quad (2)$$

Substituting $y = \frac{1}{x}$ into (1) we get

$$x^4 - x^2 - 1 = 0 \quad (3)$$

which gives $x^2 = \frac{1 \pm \sqrt{5}}{2}$. As x^2 cannot be negative we have

$$x^2 = \frac{1 + \sqrt{5}}{2} \quad (4)$$

The two square roots of the R.H.S. will give the x -coordinates of the points of intersection of the two curves. The corresponding y -coordinates will be the reciprocals of these values. So the two hyperbolas will intersect in two points, one in the first and the other in the third quadrant. We can identify them explicitly and then write down the equations of the tangents to the two curves at each point of intersection. But the question asks merely the angles between the tangents. For this it is sufficient to know merely their slopes. And these can be found from $\frac{dy}{dx}$ for both the curves by implicit differentiation. For (1), we get

$$x - y \frac{dy}{dx} = 0 \quad (5)$$

But since we are dealing with a point which lies on (2) too, we have

$$\frac{dy}{dx} = x^2 \quad (6)$$

at a point of intersection. Similarly, from (2) we have

$$y + x \frac{dy}{dx} = 0 \quad (7)$$

which, at a point (x, y) of intersection means

$$\frac{dy}{dx} = -\frac{1}{x^2} \quad (8)$$

The R.H.S. of (6) and (8) are negative reciprocals of each other. So, the tangent to the hyperbola (1) is perpendicular to that to the hyperbola (2), no matter what these points are. So, even without knowing what these points are, we know that the angle between the tangents at them is $\frac{\pi}{2}$. ■

The unusual feature of this problem is that we get the answer without calculating the slopes of the two tangents, and, indeed, without even identifying the points of intersection. Those who realise that our concern is more with the slopes of the tangents than with their points of contact will get the answer quickly. Those who spend time in identifying the two points of intersection and then writing down the equations of the tangents will waste a lot of time.

The success of the short cut hinges on two factors. First, there is no need to hastily identify the points of intersection if we merely want the slopes of the tangents to them at these points. The second factor of the short cut is that in the present problem, these tangents happen to be at right angles and so no computations are needed to calculate the angle between them. Just a glance at the expressions for their slopes is enough. If the angle were some other, say $\frac{\pi}{4}$, we would have to calculate it using the formula for the angle between two lines in terms of their slopes. So, in that sense, this problem is not sufficiently representative. Orthogonality, like parallelism, is a special relationship between two lines and deserves to get a special attention. The unfortunate part is that a numerical slip can be costly. But that is the name of the game in all MCQ's anyway.

The special importance of orthogonality comes from many sources, one of them being physics where, for example, the lines of force are always orthogonal to the equipotential lines (or surfaces, for that matter). When two curves are such that they intersect each other at right angles at every point of their intersections are said to be **orthogonal trajectories** of each other. Frequently, each curve belongs to a 1-parameter family of curves and every member of one family intersects every member of the other at right angles. As easy examples, all vertical lines are orthogonal trajectories to all horizontal lines. As a more interesting example, the family of all straight lines passing through a fixed point C is orthogonal to the family of all concentric circles centred

at C . In the present problem, the family of all hyperbolas of the form $x^2 - y^2 = a^2$ and the family of all hyperbolas of the form $xy = b^2$ are orthogonal trajectories of each other. Here a, b are the parameters of the two families.

Every 1-parameter family is characterised by a differential equation of order 1 and it is a classic application of differential equations, to find out, given such a family, the family of its orthogonal trajectories. We shall not go into it here, because differential equations are not included in the ISI Entrance syllabus. But they are included in the syllabi of comparable examinations, such as the JEE, at the HSC level. So chances are that some candidates appearing for ISI entrance already know the answer to this problem. That, of course, is no crime. But it gives an unfair advantage to one section of the candidates. Unfortunately, it is only through conventional examinations that one can realistically assess the degree of the handicap. In an MCQ, what matters is whether you answered it correctly and not how you got the answer.

Q.8 The population of a city doubles in 50 years. In how many years will it triple, under the assumption that the rate of increase is proportional to the number of inhabitants?

- | | |
|-------------------------|------------------------------------|
| (A) 75 years | (B) 100 years |
| (C) $50 \log_2 3$ years | (D) $50 \log_e(\frac{3}{2})$ years |

Answer and Comments: (C). Let t denote the time (measured in years) and let $p = p(t)$ be the population at time t . Its rate of growth is simply $\frac{dp}{dt}$ and we are given that it is proportional to p . In other words, there is some constant k such that

$$\frac{dp}{dt} = kp \tag{1}$$

This is a differential equation. Although differential equations are not a part of the ISI syllabus, this particular one is very easy to solve and chances are that many candidates have encountered similar equations in physics (e.g. radioactive decays, where the constant k is negative). The method of solving (1) is popularly called ‘separating differentials’.

That is, we rewrite it as

$$\frac{dp}{p} = kdt \quad (2)$$

knowing fully well that both the sides are meaningless, because, differentials, by themselves are meaningless. It is only a symbolic ratio of two differentials, such as the L.H.S. of (1), that has a meaning as derivative. But this approach is very common, convenient and never lands into any trouble because it has a clean back-up which will be given after the solution.

The next step is to ‘integrate’ both the sides to get

$$\log p = kt + c \quad (3)$$

where $\log$ denotes the natural logarithm $\log_e$ (more customarily denoted by $\ln$ nowadays) and c is some arbitrary constant (not necessarily related to k) whose value can be determined from the other parts of the data.

Taking exponentials of both the sides,

$$p = p(t) = c_1 e^{kt} \quad (4)$$

where $c_1 = e^c$. (Many times c_1 is denoted by c itself. After all, it is some constant and till its value is determined, it hardly matters which symbol is used for it.)

The next task is to determine c_1 and k . We are given that the population doubles in 50 years. Without loss of generality, we assume that the unit of population is so chosen that at the start, i.e. at $t = 0$, $p = 1$. Then we have

$$1 = p(0) = c_1 e^0 = c_1 \quad (5)$$

which gives

$$p(t) = e^{kt} \quad (6)$$

We are further given that when $t = 50$, $p = 2$. So,

$$2 = p(50) = e^{50k} \quad (7)$$

Hence

$$k = \frac{\log 2}{50} \quad (8)$$

Putting this into (6), we get

$$p(t) = e^{(\log 2)t/50} = (e^{\log 2})^{t/50} = 2^{t/50} \quad (9)$$

We are asked to find the value of t for which $p(t) = 3$. This is given by

$$3 = 2^{t/50} \quad (10)$$

Taking logarithms of both the sides w.r.t. 2 we get $t = 50 \log_2 3$. ■

The problem is a cakewalk for those who know even the rudiments of differential equations. Even those who don't, but have worked out problems on radioactive decay will simply start with (6) as if that was the data, because the transition from (1) to (6) is absolutely routine. This covers most of the candidates. It is only for the small underprivileged minority who have never seen a problem like this before, that we have given the solution in a pedestrian manner.

But the promised back-up to validate (2) needs to be given not only to this minority but to the vast majority as well. Instead of (2) we write

$$\frac{1}{p} \frac{dp}{dt} = k \quad (11)$$

which is perfectly legitimate. As p is a function of t , both the sides are functions of t , the R.H.S. being a constant function. If we integrate both the sides over some interval $[t_1, t_2]$ we get

$$\int_{t_1}^{t_2} \frac{1}{p} \frac{dp}{dt} dt = \int_{t_1}^{t_2} k dt \quad (12)$$

$\ln p(t)$ is an antiderivative for the integrand on the L.H.S. So

$$\ln(p(t_2)) - \ln(p(t_1)) = k(t_2 - t_1) \quad (13)$$

We are taking $p(0) = 1$. So taking t_1 as 0 and t_2 as t , we have

$$\ln(p(t)) = kt \quad (14)$$

and hence

$$p(t) = e^{kt} \tag{15}$$

which is the same as (6). No separation of differentials is needed here.

The underlying theme of this problem is **continuous growth**. It is a basic law of nature that when things, especially biological organisms grow, the rate of growth is proportional to the quantity present. This is to be expected because, other conditions being equal, between two bacterial colonies of which one is, say, thrice the size of the other, the bigger colony will grow thrice as fast as the smaller one. This law holds for radioactive decays too, where the growth is negative. It is also applied artificially to money deposited when it is given continuously compounded interest.

Mathematically, continuous growth is represented by an equation like (1). The presumption here is, of course, that p is a continuous variable, i.e. a variable which can, in theory, take any positive real number as a possible value. Mass, length, volume, are common examples of such variables. The population, on the other hand, has to be a whole number. So it is an example of what is called a **discrete variable**. Such variables grow in instantaneous jumps. When a baby is born, the population shoots up by 1. When somebody dies, it goes down by 1. And in between two such events it remains constant because there is nothing like partial birth or partial death (except possibly in phrases like ‘half dead’!). So, it may appear ridiculous to write down an equation like (1) and it will indeed be so for a family. In a family, procreation (and death too) are sporadic events because there are only a few members in a family. But in a large community, even though each individual procreates sporadically, at any given time, it is statistically safe to assume that a certain percentage of the population is procreating, even though the individuals procreating at different times may be different. And if this percentage is independent of time, then we have a situation as in this problem.

Summing up, although the population p is very much a discrete variable, for a large community we can treat it as a variable which is a function of time t , and form its derivative $\frac{dp}{dt}$. The presumption was that even though each individual procreates sporadically, collectively

they do so at different times and so at any one moment some procreation is going on. But there are many living species for which this assumption fails. For example, all cherries blossom at more or less the same time and then there is no bloom till the next spring. Similarly most of the crops are grown seasonally. So we talk of the annual food production every year. Here the natural unit of time is an year and we allow only whole multiples of it. Because of improved farming techniques, it may happen that the annual food production increases by a certain multiple of the previous year's production. In that case, if we start from some base year with food production F_0 , then the successive food productions in subsequent years, viz. $F_1, F_2, \dots$ will form a geometric progression because the ratio $\frac{F_i}{F_{i-1}}$ is constant. See Comment No. 6 of Chapter 19 for a problem where the population grows continuously but the food production grows only annually.

Similarly, in ordinary compounding of interest, simple interest at some rate, say 10% p.a. is added to the principal at regular intervals, often one year or half a year and then the total becomes the new principal for the next period. Thus, with the interest rate given above, if 1 rupee is invested then after n years the total will be $(1.1)^n$ rupees under annual compounding while under semi-annual compounding it will be $(1.05)^{2n}$ which is slightly higher. This is also a case where the growth is not continuous but sporadic.

Q.9 We define a set $\{f_1, f_2, \dots, f_n\}$ of polynomials to be a linearly dependent set if there exist real numbers $c_1, c_2, \dots, c_n$, not all zero, such that $c_1 f_1(x) + \dots + c_n f_n(x) = 0$ for all real numbers x . Which of the following is a linearly dependent set?

(A) $\{x, x^2, x^3\}$

(B) $\{x^2 - x, 2x, x^2 + 3x\}$

(C) $\{x, 2x^3, 5x^2\}$

(D) $\{x^2 - 1, 2x + 5, x^2 + 1\}$

Answer and Comments: (B). Although the question does not define it, an expression like $c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)$ is called a **linear combination** of the functions $f_1(x), f_2(x), \dots, f_n(x)$. Linearly independent set is one where the only linear combination that vanishes identically is the trivial one, i.e. one where all the coefficients are 0. The question asks for which of the given four sets of polynomials, there

is some non-trivial linear combination which vanishes identically, i.e. for all x .

Once this is understood, it is easy to rule out (A) and (C). In both of them, any linear combination of the three functions will be a cubic polynomial of the form $ax^3 + bx^2 + cx$ and hence can have at most three real roots unless it is identically zero. But that means a, b, c all vanish.

However, in (B) and (D), the polynomials are not all of different degrees. If each case, if we subtract the first polynomial from the last, we get a polynomial $g(x)$ of degree less than 2. In (B), $g(x)$ equals $4x$ which is twice the middle polynomial $2x$. So the linear combination $(-1)(x^2 - x) + (-2)(2x) + 1(x^2 + 3x)$ vanishes for every x . Hence the set $\{x^2 - x, 2x, x^2 + 3x\}$ is a linearly dependent set. ■

To complete the solution, we must dispose off (D) too. Consider a linear combination $f(x) = c_1(x^2 - 1) + c_2(2x + 5) + c_3(x^2 + 1)$. Collecting the like terms, $f(x) = (c_1 + c_3)x^2 + 2c_2x + (c_3 - c_1)$. For this to vanish identically, we must have $c_1 + c_3 = 0, c_2 = 0$ and $c_3 = c_1$. The only possible solution is $c_1 = c_2 = c_3 = 0$. (A) and (C) could also be tackled by this method, but there it was possible to arrive at the answer just from the observation that the three members in those sets are monomials of different degrees. In general the problem of deciding whether a set is linearly dependent or not boils down to checking whether a system of linear equations has a non-trivial solution. It properly belongs to a branch of mathematics called **linear algebra**.

The present problem is easy because once you understand it, it can be solved almost by inspection. So apparently the paper setters want to test the ability to understand a newly defined concept. This is commendable because many students are programmed to tackle only problems where it is clearly given what is to be done (e.g. Q.7 about the two hyperbolas). They are simply baffled when *they* have to figure out what is to be done. Unfortunately, in the present problem, this ‘new’ concept is really not new to many candidates and so they stand to gain over those who hear it for the first time.

- Q.10 A set of numbers S is said to be multiplicatively closed if $ab \in S$ whenever both $a \in S$ and $b \in S$. Let $i = \sqrt{-1}$ and ω be a non-real cube root of unity. Let $S_1 = \{a + bi : a, b \text{ are integers}\}$ and

$S_2 = \{a + b\omega : a, b \text{ are integers}\}$. Which one of the following statements is true?

- (A) Both S_1 and S_2 are multiplicatively closed.
- (B) S_1 is multiplicatively closed but S_2 is not.
- (C) S_2 is multiplicatively closed but S_1 is not.
- (D) Neither S_1 nor S_2 is multiplicatively closed.

Answer and Comments: (A). Another problem which begins by defining a new concept. But unlike the concept of linear dependence, the concept introduced here is not such a household term. So, in this respect the present problem is a more valid test of the ability to grasp something new. One wishes, however, that the candidates were not additionally taxed by the notations. In defining the concept, a, b stand for the elements of an unspecified set S . But in the specifications of the particular set S_1 and S_2 , these same symbols are used to denote elements of a particular set, viz. the set of all integers. A mature reader will not be confused by such double usage. But then, a mature reader is likely to have heard the term ‘multiplicatively closed’ anyway. Why not use the symbols x and y instead of a and b in defining a multiplicatively closed set? One wonders if it was just a lapse in paper setting or whether causing such a confusion was deliberate as a means to test maturity!

Anyway, coming to the problem, we have to check, independently, whether each of S_1 and S_2 is multiplicatively closed. Let us begin with S_1 . Take two typical elements, say $x_1 = a_1 + b_1i$ and $x_2 = a_2 + b_2i$ in S where a_1, b_1, a_2, b_2 are all integers. Then by a direct calculation and using the fact that $i^2 = -1$, we get

$$x_1x_2 = (a_1 + b_1i)(a_2 + b_2i) = (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)i \quad (1)$$

which is in S_1 because both $a_1a_2 - b_1b_2$ and $a_1b_2 + a_2b_1$ are integers.

With S_2 , we again take x_1 and x_2 as above. But this time we get

$$x_1x_2 = (a_1 + b_1\omega)(a_2 + b_2\omega) = (a_1a_2) + (a_1b_2 + a_2b_1)\omega + b_1b_2\omega^2 \quad (2)$$

Here a_1a_2 , $a_1b_2 + a_2b_1$ and b_1b_2 are all integers. But how do we tackle ω^2 ? The paper setters have been very kind here. Instead of giving

ω as $\frac{-1 + \sqrt{3}i}{2}$ or in some other frightening form, they have told us plainly that it is a non-real cube root of unity without telling us what ‘unity’ means! It is an old name for the number 1. (The old name for 0 is ‘naught’.) Today, this is probably the only place where that name is still used at least at the elementary level. So, simply stated, we are given that ω is not real but $\omega^3 = 1$. Factor $\omega^3 - 1$ as $(\omega - 1)(\omega^2 + \omega + 1)$. The first factor is non-zero since $\omega \neq 1$. So $\omega^2 + \omega + 1 = 0$ which gives

$$\omega^2 = -1 - \omega \quad (3)$$

Putting this into (2) and regrouping the terms,

$$x_1x_2 = (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1 - b_1b_2)\omega \quad (4)$$

which shows that $x_1x_2 \in S_2$. So S_2 is also multiplicatively closed. Thus (A) holds. ■

Because of familiarity with multiplication of complex numbers, (1) is very natural to arrive at. So it is easy to see that S_1 is multiplicatively closed. There is a slight problem with S_2 because of ω^2 . But the property (3) is so well-known that hardly any students will miss it. So, like the last question, this one is more a test of understanding a definition.

- Q.11 When the product of four consecutive odd positive integers is divided by 5, the set of remainder(s) is

(A) $\{0\}$ (B) $\{0, 4\}$ (C) $\{0, 2, 4\}$ (D) $\{0, 2, 3, 4\}$

Answer and Comments: (B). This is one of those problems where the choice of a convenient notation simplifies calculations. Instead of taking the integers as $x, x + 2, x + 4$ and $x + 6$ where x is an odd integer, focus on their centre, i.e. the midpoint of the two middle terms. That will be some even number, say, $2m$. Then the two odd integers neighbouring it will be $2m - 1$ and $2m + 1$, while the two at the far ends will be $2m - 3$ and $2m + 3$. So, if we denote the product by P , we have

$$P = (2m - 3)(2m - 1)(2m + 1)(2m + 3) \quad (1)$$

$$= (4m^2 - 1)(4m^2 - 9) = 16m^4 - 40m^2 + 9 \quad (2)$$

When at least one of these four integers is divisible by 5, so is P by (1). The only case left is when the midpoint $2m$ is itself a multiple of 5, and hence of the form $10k$ for some integer k . But then $m = 5k$ and putting this into (2) we see that P leaves the same remainder as 9 when divided by 5. This remainder is 4. Hence the two possible remainders are 0 and 4. ■

A good problem where only elementary number theory is involved. It is more a test of the ability to choose a convenient notation. Once that is done, the computations are elementary and can be done even mentally. The work needed could have been increased by posing the problem as a probability problem. Suppose, for example, that four consecutive odd integers from the set $\{1, 2, \dots, 99, 100\}$ are taken. (The last element is redundant, but 100 looks better than 99, so let us keep it.) Then what is the probability that their product is divisible by 5? Here we first have to identify the sample space. Again, we focus on the middle even integer $2m$. Because of the restriction, $2m$ can take any even value from 4 to 96. So there are 47 possible values of m , viz. $\{2, 3, \dots, 47, 48\}$. This set can be taken as the sample space, say S . The work above shows that the product P is divisible by 5 except when m is a multiple of 5. In S , there are 9 such multiples, ranging from 5 to 45. So the desired probability is $\frac{47 - 9}{47} = \frac{38}{47}$.

Those who love to explore will hardly stop here. There is nothing very special about the number 100 after all. What if we replace it by 200, 300 or some arbitrary even number like 478? Experimentation will show that the larger the number, the probability will come closer to $\frac{4}{5}$. This is not very surprising because an integer m taken at random is one of the five forms listed above. All these forms are regularly distributed in $\mathbb{N}$, the set of all positive integers. So assuming that each of these five cases is equally likely, since four of them are favourable, the desired probability will be $\frac{4}{5}$.

But how do we give this argument rigorously? In fact, how do we define the probability when the sample space S is infinite? Such questions are addressed in Chapter 23 on Infinitistic Probability. One method is to pose the problem as a limiting case of a problem of fini-

tistic probability. In the present problem, let p_n be the probability that the product of four consecutive odd integers taken from the set $\{1, 2, 3, \dots, 2n-1, 2n\}$ is divisible by 5. Then p_1, p_2, p_3 are undefined while $p_4 = 1$ because the only possibility, viz. $(1, 3, 5, 7)$ is a favourable one. A similar reasoning shows that p_5 and p_6 are both 1 since in each, all cases are favourable. However, for $n = 7$, out of the four possibilities $(1, 3, 5, 7), (3, 5, 7, 9), (5, 7, 9, 11)$ and $(7, 9, 11, 13)$, only the first three are favourable. So, $p_7 = \frac{3}{4}$. From the example given above, p_{50} came out as $\frac{38}{47}$.

To calculate p_n in general, we duplicate the reasoning given in the calculation of p_{50} . This time m can take the values from 2 to $n-2$. Hence the sample space S_n has $n-3$ elements. The number of unfavourable cases equals the number of multiple of 5 in the set $\{2, 3, \dots, n-2\}$. This number will be approximately $[(n-2)/5]$, the integral part of $\frac{n-2}{5}$. The exact count will differ slightly depending on n . But it will differ by at most 1. So, for large n , we may take the number of unfavourable cases to be $\frac{n-2}{5}$. Hence

$$p_n \approx \frac{4(n-2)}{5(n-3)} \quad (3)$$

So for all large n , p_n is nearly $\frac{4}{5}$. Therefore $\lim_{n \rightarrow \infty} p_n = \frac{4}{5}$. This limit is the most reasonable definition of p for an infinite set.

Problems on infinitistic probability are rare. The present one would fit in Paper 2 where p_n will have to be calculated. But it can also fit in Paper 1 because the essential idea is simple, viz. that out of the five possibilities, four are favourable.

Q.12 Consider the equation $x^2 + y^2 = 2015$ where $x \geq 0$ and $y \geq 0$. How many solutions (x, y) exist such that both x and y are non-negative integers?

- | | |
|-----------------|----------------------|
| (A) None | (B) Exactly one |
| (C) Exactly two | (D) Greater than two |

Answer and Comments: (A). Possibly following International Mathematics Olympiads, it is now becoming fashionable in annually held contests to design one problem which contains the year of the particular test as a part of the data of problem. Naturally, these figures have a decorative role. Sometimes certain features of it are relevant to the solution. But many times they are mere show pieces and any positive integer could replace them without affecting the problem.

In the present problem, the question is to see in how many ways 2015 can be expressed as a sum of squares of two non-negative integers x and y . As 2015 is not a perfect square, the possibilities where either x or y is 0 are ruled out. So the problem boils down to expressing 2015 as a sum of two positive perfect squares. The fascination for such expressions comes from a classic result **Fermat's two square theorem** (actually proved by **Gauss**) that every prime of the form $4k + 1$ can be expressed uniquely as a sum of two squares. The proof is far from trivial, But we can illustrate it with examples. For example, $5 = 2^2 + 1^2$. (To ensure uniqueness we arrange the squares in a descending order.) Similarly, $13 = 3^2 + 2^2$ and $29 = 5^2 + 2^2$. Moreover, if we have a product of two such (distinct) primes, then it can be expressed as a sum of squares in two different ways. These expressions can be obtained most easily using complex numbers. Thus we write 5 as $(2 + i)(2 - i)$ and 13 as $(3 + 2i)(3 - 2i)$. As a result

$$65 = 13 \times 5 = (3 + 2i)(3 - 2i)(2 + i)(2 - i) \quad (1)$$

If we could write the R.H.S. in the form $(a + bi)(a - bi)$, we would get $65 = a^2 + b^2$. This factor $a + bi$ can be formed by multiplying one factor of 13 and one factor of 5. Suppose we multiply $3 + 2i$ and $2 + i$. Then we get $4 + 7i$. Taking complex conjugates, the other factor is $4 - 7i$. So we get $13 = 4^2 + 7^2$ (which we can rewrite as $7^2 + 4^2$ to ensure uniqueness). But we can as well multiply the factor $3 + 2i$ of 13 with the factor $2 - i$ of 5. Then we would get $65 = 8^2 + 1^2$.

As is to be expected, if n is a product of three primes of the form $4k + 1$ each, then n can be expressed as a sum of squares in 4 different ways. (We can also include 2 as a prime factor because we can write $2 = 1^2 + 1^2 = (1 + i)(1 - i)$. But in that case, from an expression of p as $a^2 + b^2$, we do not get two distinct expressions for $2p$ as sums of squares. We get only one, viz. $(a + b)^2 + (a - b)^2$.) If n contains some

repeated prime factors, then by grouping together their even powers we can express n as r^2s where s is what is called the **square free part** of n . If s can be expressed as $a^2 + b^2$ then $n = (ra)^2 + (rb)^2$. For example, suppose $n = 360$. Then $n = 36 \times 10$ which can be expressed as $18^2 + 6^2$.

The path to the solution is now clear. We first factorise the given integer 2015. 5 is an obvious factor. It does take a little trial to factor the other factor, viz. 403. It comes out to be 13×31 . So $2015 = 5 \times 13 \times 31$. All we have to do now is to apply Fermat's theorem to each of these primes and multiply the factors using complex numbers. We already did this for the first two primes, viz. 5 and 13. But what about 31? It is a prime of the form $4k + 3$ and not $4k + 1$. So Fermat's theorem is silent. Still, there could be some way to express the particular prime 31 as a sum of two squares and then we would get the answer to our problem.

Unfortunately, such a search is futile. We have not stated it fully. But actually Fermat's theorem says that an odd prime p can be expressed as a sum of two squares *if and only if* it is of the form $4k + 1$. The case where p is of the form $4k + 3$ can be ruled out very easily and that is why it is often not included in the statement of Fermat's theorem. And, indeed, the reasoning is applicable to any number (not necessarily a prime) of the form $4k + 3$. In particular we can apply it to 2015. So, suppose we have two positive integers x and y such that

$$x^2 + y^2 = 2015 \tag{2}$$

Obviously one of x and y is even and the other odd. Suppose, without loss of generality that $x = 2m$ and $y = 2n + 1$ for some integers m, n . Then (2) becomes

$$4(m^2 + n^2 + n) = 2014 \tag{3}$$

which is impossible because the L.H.S. is divisible by 4 while the R.H.S. is not. Hence the given equation has no solutions. ■

The problem is a damp squib for those who know about Fermat's beautiful theorem. It is nowhere needed. All that is needed is that 2015 is of the form $4k + 3$. One wishes that this test were held in 1885 instead of 2015, for then we would have a factor of 29 instead of 31

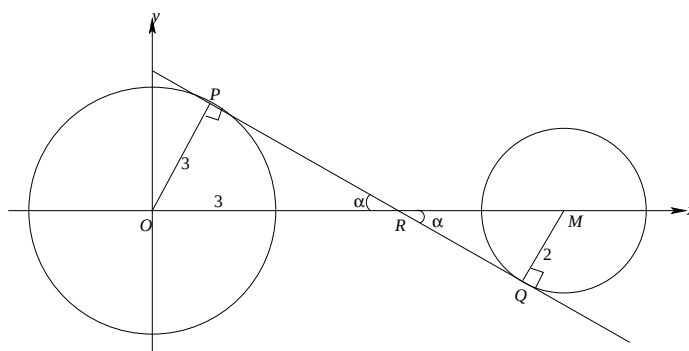
and then the given equation would have four solutions where $x > y$ and hence eight solutions in all. Now we have to wait till 2405 which is $5 \times 13 \times 37$.

One may wonder whether theorems like Fermat's have any real life applications. True, it was proved purely as an intellectual pursuit, unlike some concepts like the Bessel functions who owe their very origin to problems of physics. What is remarkable is that two centuries after its discovery, Fermat's theorem found an application in the study of vibrations of rectangular membranes! So, nothing is permanently useless in mathematics.

- Q.13 Let P be a point on the circle $x^2 + y^2 - 9 = 0$ above the x -axis, and Q be a point on the circle $x^2 + y^2 - 20x + 96 = 0$ below the x -axis such that the line joining P and Q is tangent to both these circles. Then the length of PQ is

- (A) $5\sqrt{2}$ units (B) $5\sqrt{3}$ units (C) $5\sqrt{6}$ units (D) $6\sqrt{5}$ units

Answer and Comments: (B). Call the circles as C_1 and C_2 respectively. Then C_1 has centre $O = (0, 0)$ and radius 3 while C_2 has centre $(10, 0) = M$ (say) and radius 2 as we see by completing the square. The x -axis is their common diameter.



In general, when one circle lies entirely outside the other, as in the present case, there are four common tangents to them. In two of them, their points of contact lie on the same side of their common diameter. They are called **direct tangents**. When the points of contact lie on

opposite sides of the common diameter, they are called **inverse tangents**. In this problem, PQ is an inverse tangent and we are asked to find its length.

Let PQ meet the common diameter at R . As the right angled triangles OPR and MQR are similar, we have

$$\frac{OR}{RM} = \frac{OP}{MQ} = \frac{3}{2} \quad (1)$$

But $OR + RM = OM = 10$. So we get

$$OR = 6, RM = 4 \quad (2)$$

From the right angled triangles OPR and MQR , we get

$$\begin{aligned} PQ &= PR + RQ \\ &= \sqrt{OR^2 - OP^2} + \sqrt{RM^2 - MQ^2} \\ &= \sqrt{36 - 9} + \sqrt{16 - 4} \\ &= 3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3} \end{aligned} \quad (3)$$

Hence (B) is correct. ■

A nice, simple geometry problem. But now that the cushion of subjective Paper 2 is available, this would fit more appropriately in Paper 2 where the candidates are asked not only to find the length of PQ , but to give a construction for it. In the present problem, the construction would begin by finding the point R which divides OM in the ratio of the radii. This is a relatively simple construction based on similar triangles. The next task is to draw tangents from R . For this we first draw a semi-circle with OR as a diameter and letting P be its point of intersection with the circle C_1 . Once we find P , there is no need to find Q as the line PR will touch C_2 too. See the variety of results needed in geometric constructions. This most challenging aspect of geometry has simply no room in an MCQ. It is high time ISI revives it through Paper 2.

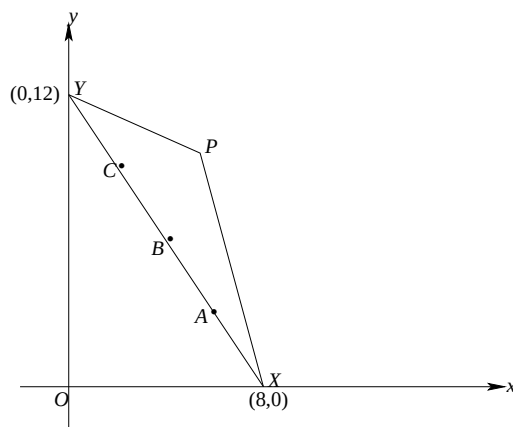
Q.14 Let $S = \{(x, y) : x, y \text{ are positive integers}\}$ viewed as a subset of the plane. For every point P in S , let d_P denote the sum of the distances

from P to the point $(8, 0)$ and the point $(0, 12)$. The number of points P in S such that d_P is the least among all elements in the set S , is

- (A) 0 (B) 3 (C) 8 (D) 1

Answer and Comments: (B). Call the points $(8, 0)$ and $(0, 12)$ as X and Y respectively. For any point P in the plane, d_P is defined by

$$d_P = PX + PY \quad (1)$$



We can get a formula for d_P as a function of the coordinates x and y of P . But that is hardly necessary here. From the triangle PXY , we see that the sum of the sides PX and PY is at least the third side XY with equality holding if and only if P lies on the segment XY . This is the most elementary and natural form of what is called the **triangle inequality**. This is in fact, where the inequality gets its name from although it has many sophisticated generalisations.

So, for d_P to be minimum, P must lie on the segment XY . Moreover, P is required to have only integer coordinates. The equation of the line XY (in the intercepts form) is

$$\frac{x}{8} + \frac{y}{12} = 1 \quad (2)$$

So the problem boils down to finding all solutions of (2) in which both x, y are positive integers. The points X and Y are not allowed. But by

trial we can put $x = 6, 4$ and 2 and correspondingly get $y = 3, 6$ and 9 . The resulting three points A, B, C are shown in the figure. ■

Once the idea that P must lie on the segment XY strikes, the rest of the work is very simple. So this is a problem well-suited as an MCQ. Writing the equation of the line XY in the intercepts form is somewhat unusual. But that is the form that suits here best. For those who insist upon rigour rather than inspection, (2) can be written as

$$3x + 2y = 24 \quad (3)$$

So,

$$3x = 2(12 - y) \quad (4)$$

So $3x$ and hence x is even. Further $0 < x < 8$. So the possible values of x are $2, 4$ and 6 .

Q.15 Let A, B and C be the angles of a triangle. Suppose that $\tan A$ and $\tan B$ are the roots of the equation $x^2 - 8x + 5 = 0$. Then $\cos^2 C - 8 \cos C \sin C + 5 \sin^2 C$ equals

- (A) -1 (B) 0 (C) 1 (D) 2

Answer and Comments: (C). We are not given $\tan A$ and $\tan B$ individually. Nor do we need them. We are given that they are the roots of the quadratic $x^2 - 8x + 5 = 0$. From this we get

$$\tan A + \tan B = 8 \quad (1)$$

$$\text{and } \tan A \tan B = 5 \quad (2)$$

And this is good enough to get $\tan C$. For

$$\begin{aligned} \tan C &= \tan(\pi - A - B) \\ &= -\tan(A + B) \\ &= \frac{\tan A + \tan B}{\tan A \tan B - 1} \end{aligned} \quad (3)$$

$$= \frac{8}{5 - 1} = 2 \quad (4)$$

(This could have also been obtained from the standard identity

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C \quad (5)$$

But that is really not a different method because the standard proof of this identity is by going through (3).)

Knowing that $\tan C = 2$, and keeping in mind that $\sin C > 0$ as C is the angle of a triangle, we get

$$\sin C = \frac{2}{\sqrt{5}} \quad (6)$$

$$\text{and } \cos C = \frac{1}{\sqrt{5}} \quad (7)$$

A stright substitution gives

$$\cos^2 C - 8 \cos C \sin C + 5 \sin^2 C = \frac{1}{5} - \frac{16}{5} + 4 = 1 \quad (8)$$

So (C) is correct. ■

An absolutely straightforward problem which is a striaght application of (5). But a slight twist is given by giving $\tan A, \tan B$ as the roots of a quadratic.

Q.16 Let $A = \{a_1, a_2, \dots, a_{10}\}$ and $B = \{1, 2\}$. The number of functions $f : A \longrightarrow B$ for which the sum $f(a_1) + f(a_2) + \dots + f(a_{10})$ is even is:

- (A) 128 (B) 256 (C) 512 (D) 768

Answer and Comments: The sum of 10 numbers is even if and only if the number of odd terms in it is even. In the present case, this means that the function f takes the value 1 at an even number of points in A . Equivalently,

$$|f^{-1}(\{1\})| = 2r \quad (1)$$

for some $r = 0, 1, 2, 3, 4$ and 5 . (Here $|X|$ denotes the number of elements in a set X .) Since f takes only two possible values, viz. 1 and 2, for a fixed r , it is uniquely determined by this subset $f^{-1}(\{1\})$. There

are $\binom{10}{2r}$ such subsets of A . So, the total number of functions of the type we want is the sum, say S ,

$$S = \sum_{r=0}^5 \binom{10}{2r} \quad (2)$$

There are several ways to evaluate this sum. The brute force method is to calculate each term and add. Considering that the numbers involved are small, this is not such an impracticable idea. We can also use the binomial identity

$$\binom{n}{k} = \binom{n}{n-k} \quad (3)$$

to shorten the calculation because we need to calculate only upto $r = 2$ and double the count. Thus

$$\begin{aligned} S &= 2 \left[\binom{10}{0} + \binom{10}{2} + \binom{10}{4} \right] \\ &= 2 \left(1 + 45 + \frac{10 \times 9 \times 8 \times 7}{24} \right) \\ &= 2(46 + 210) = 512. \end{aligned} \quad (4)$$

Although the solution is complete, a perceptive reader will hardly fail to note that the answer is exactly half of 1024, which is the number of all functions from A to B . This can hardly be a coincidence. So, one is led to inquire if a more general result is true, viz. for every n

$$\sum_{r=0}^{\infty} \binom{n}{2r} = 2^{n-1} \quad (5)$$

Note that the sum on the L.H.S. is ostensibly infinite. But in reality, for each fixed n , $\binom{n}{2r} = 0$ if $r > n/2$ and so it is only a finite sum.

The advantage gained by writing it as an infinite sum is that we do not have to worry about the upper limit of summation. Otherwise, we have to let r go from 0 to $n/2$ if n is even and from 0 to $(n-1)/2$ if n is odd. And so we would have to display two separate formulas, one for even n and the other for odd n .

When n is large, the brute force method which worked in our example will be horrendous and in any case it will not be applicable for a variable n . So, we look for some other method to prove (5). One such method is provided by the binomial theorem. Although the theorem holds for the expansion of the n -th power of the sum of any two numbers, the case which is most needed in combinatorial problems is one where one of these numbers is 1 and the other a variable x . So we state only that.

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k \quad (6)$$

$$= \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n \quad (7)$$

Even though (6) is an ostensibly infinite sum, in reality it is the finite sum (7) and so we are free to give any real (or even complex) value. (For infinite sums, we have to worry about convergence and that restricts the values we can give to x .)

In particular, if we put $x = 1$ we get

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{k} + \dots + \binom{n}{n} = (1+1)^n = 2^n \quad (8)$$

On the other hand, if we put $x = -1$, we get

$$\binom{n}{0} - \binom{n}{1} + \dots + (-1)^k \binom{n}{k} + \dots + (-1)^n \binom{n}{n} = (1-1)^n = 0 \quad (9)$$

If we add (8) and (9), terms of the form $\binom{n}{2r}$ get added while those of the form $\binom{n}{2r+1}$ get cancelled. So we get (5) and the special case $n = 10$ gives the solution to our problem as $2^9 = 512$.

So far we have given two solutions, one based on a direct computation and the other on the binomial theorem. The paper-setters have given us a hint for yet another solution. Let $T(f)$ denote

$$T(f) = f(a_1) + f(a_2) + \dots + f(a_{10}) \quad (10)$$

We want the number of those functions f for which $T(f)$ is even. We translated this to mean that $|f^{-1}\{1\}|$ is even. This reduced the problem to finding the number of subsets of A containing an even number of elements. That led to (2) which we then evaluated by two methods.

This is the norm in most problems. We often have to reduce the problem to a more manageable one. For example, in Q.14 we translated the condition on the point P to mean that it lies on the segment XY , and then we counted such points whose coordinates are positive integers.

But in the present problem, the format of the problem itself provides a clue to the method. Let us keep $T(f)$ as it is and see what happens if we change just one term in it. Every term is either 1 or 2. So no matter which term we change, $T(f)$ will alter its parity (i.e. change from an even number to odd or from an odd number to an even one.) To put this systematically, let F denote the set of all functions from A to B . Then F has 2^{10} elements. Now, let F_0 be the set of those functions in F for which $T(f)$ is even. In symbols,

$$F_0 = \{f \in F : T(f) \text{ is even}\} \quad (11)$$

The problem asks us to find $|F_0|$. Consider the complement F'_0 of F_0 in F . Now define a function $*$: $F \rightarrow F$ by $*(f) = f^*$ where $f^* : A \rightarrow B$ is defined by

$$f^*(a_i) = \begin{cases} 2 & \text{if } i = 1 \text{ and } f(a_1) = 1 \\ 1 & \text{if } i = 1 \text{ and } f(a_1) = 2 \\ f(a_i) & \text{if } i > 1 \end{cases} \quad (12)$$

Verbally, f^* differs from f at a_1 , but agrees with it for all other i . Evidently, $*$ is a bijection. In fact it is its own inverse because $(f^*)^* = f$ for all $f \in F$. As noted earlier, for every $f \in F$, $T(f)$ and $T(f^*)$ always have opposite parities. So $f \in F_0$ if and only if $f^* \in F'_0$. In other words $*$ maps F_0 onto F'_0 . As $*$ is a bijection, F'_0 has the same number of elements as F_0 . But together they have 2^{10} elements. So each one of them has half as many, i.e. $2^9 = 512$ elements.

The problem is remarkable in that its apparently clumsy formulation itself gives a hint to its solution. The paper setters have been kind enough to keep the size of A small enough to permit a solution by direct

counting too. But once again the identity (5) is so well known (see, for example, Comment No. 14, of Chapter 5), that many candidates are likely to know it already. For them the problem is very easy and the imaginative formulation of it is lost on them.

The last solution is reminiscent of complementary counting, because of the involvement of the complement F'_0 of F_0 in it. But it is not quite the same. In a genuine case of **complementary counting**, we have to find $|S|$ for a given set S . We then recognise S as a subset of a suitable set, say T (which may or may not be given) and calculate $|T|$ and $|S'|$ where S' is the complement of S in T (also often written as $T - S$). Then we get

$$|S| = |T| - |S'| \quad (13)$$

whose proof is sheer common sense. This is especially convenient in finding complementary probability, for example, in finding out the probability that in a given class of 50 students, at least two share a common birthday. (See Comment No. 4 of Chapter 22. See also Comment No. 6 of Chapter 1 for an elegant application of complementary counting due to **Andre**.)

In other words, in complementary counting, we genuinely compute $|T|$ for the ambient set T and also $|S'|$. But in the present problem, we are counting neither $|F_0|$ (which is asked) nor $|F'_0|$. Instead, we have shown by a clever bijection, that their counts are equal and hence each is half of $|F_0 \cup F'_0|$ which is easy to calculate. In this respect the problem is of the same spirit as Q.21 of the ISI 2014 Paper 1, where the area of a subset R of a disc of radius 10 was to be found. Far more common occurrences of this trick are those where a given definite integral, say I , is converted to some other integral by a suitable substitution in such a way that $2I$ is easier to calculate directly. (See again Q.5 of ISI 2014 for an example. Also see Q.29 of the same paper for the analogue of complementary counting in evaluation of integrals.)

$$\text{Q.17 Define } \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = (x + 1) \operatorname{sgn}(x^2 - 1)$ where $\mathbb{R}$ is the set of real numbers. Then the number of discontinuities

of f is:

- (A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (B). The function defined in the problem is called the **signum function** since it signifies the sign of a number. It is hardly necessary to draw its graph. It is a prototype of a function with a jump discontinuity and is important in applications to handle what are called **impulses**, i.e. quantities which suddenly reverse their signs, as for example the velocity of a cricket ball when it is hit by a bat.

Coming to the problem, let us write the data as

$$f(x) = g(x)h(x) \quad (1)$$

where $g(x) = x + 1$ and $h(x) = \operatorname{sgn}(x^2 - 1)$. The first function $g(x)$ is continuous everywhere. As for the second, the signum function is discontinuous at 0 and hence $h(x)$ will be discontinuous when $x^2 - 1 = 0$, i.e. when $x = \pm 1$. Since the product of two functions is continuous at any point where both the functions are continuous, we see that $f(x)$ is continuous at all points other than ± 1 .

So, it is tempting to think that number of discontinuities of f is 2. BUT that is a blind application of the result about continuity of a product of two functions. Although it does say that $g(x)h(x)$ is continuous at x_0 if both g and h are continuous at x_0 , it says nothing as to what happens when one of these functions, say $h(x)$ is *not* continuous at x_0 . The other factor may ‘mask’ or ‘cure’ this discontinuity. An extreme example is when $g(x)$ is the identically zero function. More generally, if $g(x_0) = 0$ then we have to examine the discontinuity of $f(x)$ at x_0 more carefully. If $g(x_0) \neq 0$, then we can safely say that discontinuity of h at x_0 implies that of f , because in this case $g(x) \neq 0$ in some neighbourhood of x_0 and so we can write $f(x) = \frac{h(x)}{g(x)}$ which would be continuous at x_0 , if h, g both are.

This takes care of the point 1, because $g(1) = 2 \neq 0$. Hence $f(x)$ is discontinuous at 1. We now only have to check if $h(x)$ is discontinuous at -1 , where g vanishes. Then $g(-1) = 0$. Also, $g(x) \rightarrow 0$ as $x \rightarrow -1$. Since the factor $h(x)$ is bounded in a neighbourhood of -1 (in fact

everywhere), $g(x)h(x) \rightarrow 0$ as $x \rightarrow -1$. So $f(x)$ is continuous at -1 . Thus it has only one point of discontinuity, viz. 1. ■

When $g(x_0) = 0$, it may sometimes happen that $f(x)$ is discontinuous at x_0 , as we see by taking $g(x) = x$, $h(x) = \frac{1}{x^2}$ and $x_0 = 0$. Here h is not bounded in a neighbourhood of x_0 and so we need a stronger medicine to cure its discontinuity. For example, $g(x) = x^2$, or $g(x) = \sin^2 x$ will work.

Problems of this type where the discontinuity of a factor at a point gets cured because of the other factor which vanishes at it have been asked in JEE before. Interestingly, if $f(x) = g(x) + h(x)$ then continuity of g and discontinuity of h at x_0 always implies that of $f(x)$ at x_0 as we see by writing $h(x) = f(x) - g(x)$. This was asked in JEE 1987.

Q.18 Suppose X is a subset of real numbers and $f : X \rightarrow X$ is a bijection (that is, one-to-one and onto) satisfying $f(x) > x$ for all $x \in X$. Then X cannot be:

- (A) the set of integers (B) the set of positive integers
(C) the set of positive real numbers (D) the set of real numbers

Answer and Comments: (B). If X has a smallest element, say x_0 , then a function $f : X \rightarrow X$ of the given type cannot exist. If it does, then, being a bijection there exists some $x_1 \in X$ such that $f(x_1) = x_0$. But then $x_1 > x_0$ (as x_0 is the least element) while $f(x_1) > x_1$ which means $x_0 > x_1$, a contradiction. Hence in (B), no such f can exist. However, for each of the remaining three subsets, such a function exists. For (A) and (D), take $f(x) = x + 1$ while for (C) take $f(x) = 2x$. ■

A very simple problem once the idea of a least element strikes.

Q.19 The set of real numbers x satisfying the inequality

$$\frac{4x^2}{(1 - \sqrt{1 + 2x})^2} < 2x + 9$$

is

$$\begin{array}{ll}
\text{(A)} \left[-\frac{1}{2}, 0\right) \cup \left(0, \frac{45}{8}\right) & \text{(B)} \left[-\frac{1}{2}, 0\right) \cup \left(\frac{45}{8}, \infty\right) \\
\text{(C)} \left[-\frac{1}{2}, 0\right) \cup (0, \infty) & \text{(D)} \left(0, \frac{45}{8}\right) \cup \left(\frac{45}{8}, \infty\right)
\end{array}$$

Answer and Comments: (A). The solution sets of inequalities rarely come out to be nice single intervals. They are often the disjoint unions of several intervals, some of which may be unbounded. Also one has to keep track of whether the end points of these intervals are in the solution set.

As a result, in such problems, it is a good idea to keep track of whatever inequalities are implied by certain features of the data. Later on some of these inequalities may turn out to be redundant because some stronger inequalities are implied. But it is better to do the duplication than to miss something vital.

In the present problem, the L.H.S., being a square, is non-negative and so the R.H.S. is positive. This gives us one inequality on x , viz.

$$x > -\frac{9}{2} \quad (1)$$

Next, the very occurrence of $\sqrt{1+2x}$ implies $1+2x \geq 0$, i.e.

$$x \geq -\frac{1}{2} \quad (2)$$

This renders the earlier inequality redundant and so we may safely forget it. (But clever paper setters sometimes use such superficial implications to design fake options which are tempting to a hasty candidate.)

Going further, let us get rid of the ugly radicals by making a substitution

$$u = \sqrt{1+2x} \quad (3)$$

Then $2x = u^2 - 1$ and so the given inequality translates as

$$\frac{(u^2 - 1)^2}{(1 - u)^2} < u^2 + 8 \quad (4)$$

Note that the L.H.S. is undefined when $u = 1$, i.e. when $2x = 0$. So we have to exclude 0 from the solution set. For $u \neq 1$ we cancel the factor $(u - 1)^2$ from the numerator and the denominator to reduce (4) to

$$(u + 1)^2 < u^2 + 8 \quad (5)$$

which further simplifies to

$$u < \frac{7}{2} \quad (6)$$

Translated in terms of x , this means $\sqrt{1+2x} < \frac{7}{2}$. As we are taking only the positive square root, this is equivalent to

$$1+2x < \frac{49}{4} \quad (7)$$

which gives

$$x < \frac{45}{8} \quad (8)$$

So the solution set is obtained by combining (2) and (8) and excluding the point 0. Hence it is $[-\frac{1}{2}, 0) \cup (0, \frac{45}{8})$. ■

A simple problem once the idea of the right substitution (3) strikes. Even without it the problem can be solved by expanding the denominator of the L.H.S. as $4(1-\sqrt{1+2x})^2$ and then rationalising the fraction. But that will be a little complicated and prone to a numerical slip. As suggested earlier, some fake options could have involved $-\frac{9}{2}$.

- Q.20 Let $ABCDEFGHIJ$ be a 10-digit number, where all the digits are distinct. Further, $A > B > C$, $A + B + C = 9$, $D > E > F > G$ are consecutive odd numbers and $H > I > J$ are consecutive even numbers. Then A is

- (A) 8 (B) 7 (C) 6 (D) 5

Answer and Comments: (A). The only odd digits are 1, 3, 5, 7 and 9. Since $A + B + C = 9$, an odd number, either exactly one or all three of A, B, C are odd. The latter possibility is ruled out because then only two odd digits would be left and the third condition will be violated. So, only one of A, B, C is odd. The only way we can have four consecutive odd digits in a descending order is $9 > 7 > 5 > 3$ and $7 > 5 > 3 > 1$. The latter possibility means that the missing odd digit, viz. 9 is in A, B, C . But then the remaining two of these three will add to 0. Hence both will be 0, which is not allowed as all the letters represent

distinct digits. So we conclude that $D = 9, E = 7, F = 5, G = 3$ and 1 is among A, B, C . So the remaining two add to 8.

Next we focus on the last condition, viz., that H, I, J are consecutive even numbers with $H > I > J$. This time there are three possibilities, (i) $8 > 6 > 4$, (ii) $6 > 4 > 2$ and (iii) $4 > 2 > 0$. In each case, the missing two even digits are in A, B, C and add to 8. This rules out (i) and (iii). So we are left with (ii). Now the missing two even digits, viz. 8 and 0 are in A, B, C . Hence A, B, C are 1, 0 and 8. As they are given to be in a descending order, we have $A=8$. ■

Such cute little problems which any layman can do belong more appropriately to newspaper riddles. But since they demand deductive reasoning, and deductive reasoning is the soul of mathematics, once in a while it is all right to include such a problem. The reasoning is non-trivial and will occupy considerable length to write down fully as we have done above. But it does not take much time to conceive it mentally. So such a question is ideally suited as an MCQ. Although not asked, the information determines the number uniquely as 8109753642. So some further exploration, e.g. its divisibility by some chosen primes can be added to make it suitable for paper 2 as well.

- Q.21 Let $A = \{(a, b, c) : a, b, c \text{ are prime numbers, } a < b < c, a + b + c = 30\}$. The number of elements in A is

(A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (C). In essence the question asks in how many ways 30 can be expressed as a sum of three distinct primes. Since 30 is an even number, not all three of these primes can be odd. But the only even prime is 2. So we must have $a = 2$ and hence $b + c = 28$. Since $b < c$, the only possible values b can take are 3, 5, 7, 11 and 13. It is only in two of these, viz. 5 and 11 that $28 - b$ is also a prime. So there are only two members in A , viz. $(2, 5, 23)$ and $(2, 11, 17)$. ■

A shockingly disappointing problem, especially coming after the last one. Here the data involves primes which many laymen may not know. But what we do with these primes is utterly trivial. Nothing more than listing all primes upto 30 is involved. The last problem had

the naivette of a folklore tune hummed by a casual labourer to ease his toil. The present one is like a singer making an elaborate set-up with accompanying instruments and then giving an absolutely lousy performance.

Q.22 Let $f(x) = \begin{cases} \frac{|\sin x|}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Then $\int_{-1}^1 f(x)dx$ is equal to

- (A) $\frac{2\pi}{3}$ (B) $\frac{3\pi}{8}$ (C) $-\frac{\pi}{4}$ (D) 0

Answer and Comments: (D). Note that the integrand is not continuous at 0, because its limit as $x \rightarrow 0$ from the right is 1 while its limit from the left is -1 . But as long as a function is bounded, its discontinuity at a finite number of points changes neither its integrability, nor the value of the integral. So, we simply ignore 0. For every $x \neq 0$, $f(-x) = -f(x)$. So f is an odd function. Therefore, its integral over an interval $[-1, 1]$ which is symmetric about 0 is 0. ■

It is quite well known that the integral of an odd function over an interval which is symmetric about 0 vanishes. So surely testing this knowledge could not be the purpose of the question. Apparently, the problem is designed to test only the knowledge that discontinuity of a function at a finite number of points does not affect its integral.

Q.23 Let $f : (0, 2) \cup (4, 6) \rightarrow \mathbb{R}$ be a differentiable function. Suppose also that $f'(x) = 1$ for all $x \in (0, 2) \cup (4, 6)$. Which of the following is ALWAYS true?

- (A) f is increasing
 (B) f is one-to-one
 (C) $f(x) = x$ for all $x \in (0, 2) \cup (4, 6)$
 (D) $f(5.5) - f(4.5) = f(1.5) - f(0.5)$

Answer and Comments: (D). If the domain of f were a single interval, say (a, b) , then the hypothesis $f'(x) = 1$ for all $x \in (a, b)$ would imply

$$f(x) = x + \alpha \tag{1}$$

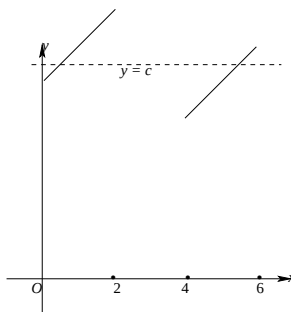
for some constant α . (For a proof, either solve the differential equation $\frac{dy}{dx} = 1$ or simply apply the LMVT to the function $g(x) = f(x) - x$ whose derivative vanishes identically.) In that case, both (A) and (B) would be true.

But in the present problem the domain is a union of two disjoint intervals, viz. $(0, 2)$ and $(4, 6)$. We can still apply the reasoning above to get

$$f(x) = x + \alpha \text{ for all } x \in (0, 2) \quad (2)$$

$$f(x) = x + \beta \text{ for all } x \in (4, 6) \quad (3)$$

But the constants need not be the same. By adjusting these constants we can give examples where (A), (B), (C) fail. For example, take $\alpha = 5$ and $\beta = 0$. Then even though $1 < 5$, we have $f(1) = 6$ and $f(5) = 5$. So although f is increasing on each of $(0, 2)$ and $(4, 6)$, it is not so on their union. (See the figure below.) Also (B) is false as we see from the line $y = c$ which cuts the graph of f in two distinct points. (C) is false for $x \in (0, 2)$. That leaves only (D) as the correct option.



For an honest proof, from (2) we have

$$f(5.5) - f(4.5) = 5.5 + \alpha - (4.5 + \alpha) = 1 \quad (4)$$

and similarly,

$$f(1.5) - f(0.5) = 1.5 + \beta - (0.5 + \beta) = 1 \quad (5)$$

So (D) is correct. ■

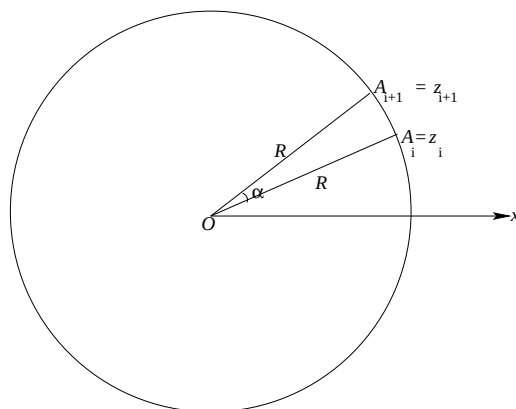
A simple problem. Those who apply Lagrange MVT blindly (i.e. without realising that α and β may be different) will be confused because both (A) and (B) would be true. Perhaps the idea was only to catch them.

Q.24 Consider 50 evenly placed points on a circle with centre at the origin and radius R such that the arc length between any two consecutive points is the same. The complex numbers represented by these points form

- (A) an arithmetic progression with common difference $(\cos(\frac{2\pi}{50}) + i \sin(\frac{2\pi}{50}))$
- (B) an arithmetic progression with common difference $(R \cos(\frac{2\pi}{50}) + i R \sin(\frac{2\pi}{50}))$
- (C) a geometric progression with common ratio $(\cos(\frac{2\pi}{50}) + i \sin(\frac{2\pi}{50}))$
- (D) a geometric progression with common ratio $(R \cos(\frac{2\pi}{50}) + i R \sin(\frac{2\pi}{50}))$

Answer and Comments: (C). There is some duplication in the data. Having said that the points are evenly placed, there was no need to say that the arc length between any two consecutive points is the same. Perhaps that was done keeping in mind the possibility that some candidates, weak in colloquial English, might not understand what is meant by ‘evenly placed’. If so, such concern is commendable. But the whole thing could have been avoided by saying that the points are the vertices of a regular 50-gon inscribed in a circle of radius R with its centre at the origin. The term ‘regular polygon’ is of a common usage. In 2014 itself, there were two questions about regular polygons, one (Q.5) about a regular pentagon and the other (Q.19) about a regular polygon with 18 vertices. The present problem could have been worded more succinctly to say that $A_1, A_2, \dots, A_{50}$ are the vertices of a regular 50-gon inscribed in a circle of radius R centred at the origin. While it is understood that the paper setting team may be different every year, one wishes some consistency of convention and diction.

Anyway, coming to the problem, even though the data is geometric it is difficult to draw a 50-gon. Nor is it necessary. What matters is only that for any two consecutive vertices A_i and A_{i+1} , the angle $\angle A_{i+1}OA_i$ (shown by the figure below) equals $\frac{2\pi}{50}$.



Let $z_1, z_2, \dots, z_{50}$ be the complex numbers represented by $A_1, A_2, \dots, A_{50}$ respectively. The problem ought to have specified whether the points are taken in a clockwise or anticlockwise manner, because the answer will change depending on that. We take them in the anticlockwise manner. In that case, since z_{i+1} is obtained from z_i by an anticlockwise rotation through $\frac{2\pi}{50}$, we get

$$\begin{aligned} z_{i+1} &= e^{2\pi i/50} z_i \\ &= \left(\cos\left(\frac{2\pi}{50}\right) + i \sin\left(\frac{2\pi}{50}\right) \right) z_i \end{aligned} \quad (1)$$

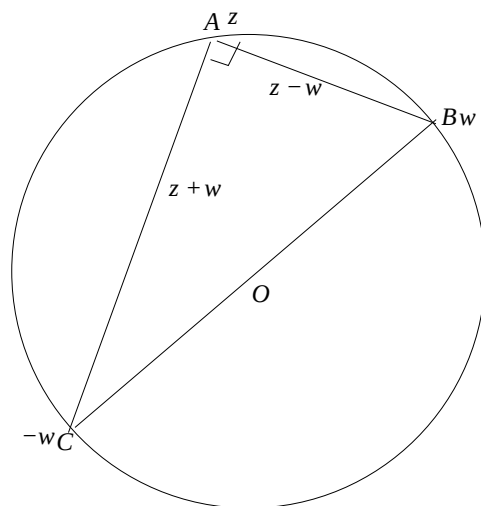
which shows that $z_1, z_2, \dots, z_{50}$ form a geometric progression with common ratio $\left(\cos\left(\frac{2\pi}{50}\right) + i \sin\left(\frac{2\pi}{50}\right) \right)$. ■

All that the problem takes is the Argand diagram representation of complex numbers. Yet another problem in a long list of trivial problems. The paper setters have not even bothered to simplify $\frac{2\pi}{50}$ as $\frac{\pi}{25}$. Had they done so, the candidate would have to do at least an iota of thinking to figure out its genesis.

Q.25 Given two complex numbers z, w with unit modulus (i.e., $|z| = |w| = 1$), which of the following statements will ALWAYS be correct?

- (A) $|z + w| < \sqrt{2}$ and $|z - w| < \sqrt{2}$
- (B) $|z + w| < \sqrt{2}$ and $|z - w| \geq \sqrt{2}$
- (C) $|z + w| \geq \sqrt{2}$ or $|z - w| \geq \sqrt{2}$
- (D) $|z + w| < \sqrt{2}$ or $|z - w| < \sqrt{2}$

Answer and Comments: (C). This problem is also based on the geometric representation. Represent z and w by points A and B on the unit circle. Then the point C diametrically opposite to B is $-w$ as shown in the figure below.



The triangle BAC is right angled at A and the sides BA and CA represent the complex numbers $z - w$ and $z - (-w) = z + w$ respectively. If we denote their absolute values by u and v respectively, then from $\triangle BAC$, we have

$$u^2 + v^2 = 4 \quad (1)$$

For this to hold, the larger of u and v must be at least $\sqrt{2}$. Since we do not know which of them is bigger, neither (A) nor (B) is *always* true. But (C) is always true. (D) would also have been true if the inequalities were not strict. ■

This problem is better than the last one but comes nowhere close to Q.9 or Q.26 of ISI 2014 which also were problems about absolute values of differences of complex numbers.

Q.26 The number of points in the region $\{(x, y) : x^2 + y^2 \leq 4\}$ satisfying $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ is

- (A) 1 (B) 2 (C) 3 (D) 4

Answer and Comments: (D). Since $\tan^4 x$ and $\cot^4 x$ are positive and reciprocals of each other, by the A.M.-G.M. inequality, the L.H.S. is at least 3. So the only way equality can hold is if each side equals 3, which gives $\tan^4 x = \cot^4 x$ and $\sin^2 y = 1$. The former implies that x is an odd multiple of $\frac{\pi}{4}$ while the latter implies that y is an odd multiple of $\frac{\pi}{2}$. Since $x^2 + y^2 \leq 4$, there are only four possibilities for (x, y) , viz. $(\pm\frac{\pi}{4}, \pm\frac{\pi}{2})$. (We need here that $5\pi^2 \leq 64$ which is true since $\pi < 3.2$ and so $\pi^2 < 10.24$.) Hence there are four points in the region satisfying the equation. ■

The solution to Q.5 was also based on the A.M.-G.M. inequality. There we wanted to minimise a sum and so the application of the A.M.-G.M. inequality was direct and obvious. In the present problem it is a little subtle because the problem is posed as a problem of solving an equation, created by matching the extreme case of the A.M.-G.M. inequality with the extreme value of $3\sin^2 y$. Such problems are often critically solvable. Suppose, for example, that we consider the equation

$$\tan^4 x + \cot^4 x + 1 = k \sin^2 y \quad (2)$$

Then for $k < 3$ there is no solution while for $k > 3$ there are infinitely many solutions and there is no easy way to identify them. So, it is only for the critical value 3 of the parameter k that there are finitely many solutions. In this sense the problem is ‘accidentally’ solvable

There was little point in asking Q.5 in presence of the present question.

- Q.27 If all the roots of the equation $x^4 - 8x^3 + ax^2 + bx + 16 = 0$ are positive, then $a + b$
- (A) must be -8
 - (B) can be any number strictly between -16 and -8
 - (C) must be -16
 - (D) can be any number strictly between -8 and 0

Answer and Comments: (A). There are no easy methods for solving a fourth degree equation. So we look for some other methods. Suppose that the roots of the given equation, viz.

$$x^4 - 8x^3 + ax^2 + bx + 16 = 0 \quad (1)$$

are $\alpha, \beta, \gamma, \delta$. Then there are the following well known identities (obtained by factoring the polynomial as $(x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$ and equating the coefficients of the like powers).

$$\alpha + \beta + \gamma + \delta = 8 \quad (2)$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = a \quad (3)$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -b \quad (4)$$

$$\text{and } \alpha\beta\gamma\delta = 16 \quad (5)$$

By (2), the A.M. of $\alpha, \beta, \gamma, \delta$ is 2. But by (5), their G.M. is also 2. So, once again by the A.M.-G.M. inequality, $\alpha, \beta, \gamma, \delta$ must all be the same. So by (2) (or by (5)) each of them equals 2. A straight substitution into (3) and (4) gives $a = 24$ and $b = -32$. So $a + b = -8$. ■

Yet another problem based on the A.M.-G.M. inequality, or rather, on its degenerate case. If the constant term in (1) were 15 (or some other positive number less than 16), instead of 16, there would be infinitely many possible values for $a + b$. On the other hand, if it were bigger than 16, there would be no solutions as the A.M.-G.M. inequality would be violated.

There are also other examples in mathematics where there is a trichotomy of possibilities, viz. no solution, a unique (or only a finite number of) solution or infinitely many solutions. Most well known is a system of n linear equations in n unknowns. If Δ , the determinant of the coefficient matrix, is 0 then there is either no solution or there are infinitely many solutions. If $\Delta \neq 0$, then there is a unique solution. But such problems cannot be called ‘accidentally solvable’, because Δ can be any real number and so its non-vanishing is the norm and not an accident. It is the vanishing of Δ that is an accident.

So, like the last problem, this one is also accidentally solvable. What such problems test is not so much mathematical ability, but rather the ability to recognise that somebody is pulling a trick on you. So these problems are good for recreational activities or for sportive contests. Whether they deserve to be asked (and asked so repeatedly) in serious examinations meant to admit students for a prestigious academic career is debatable. True mathematical aptitude goes much beyond pulling smart tricks to solve problems tailor made for them.

Q.28 Let O denote the origin and A, B denote respectively the points $(-10, 0)$ and $(7, 0)$ on the x -axis. For how many points P on the y -axis will the lengths of all the line segments PA, PO and PB be positive integers?

- (A) 0 (B) 2 (C) 4 (D) infinite

Answer and Comments: (B). Let $P = (0, y)$. Then

$$PA = \sqrt{y^2 + 100} \quad (1)$$

$$PB = \sqrt{y^2 + 49} \quad (2)$$

So the problem reduces to finding for how many integers y , $y^2 + 100$ and $y^2 + 49$ are perfect squares, say u^2 and v^2 respectively. Clearly, $u > v$. Subtracting and factoring,

$$(u + v)(u - v) = 51 \quad (3)$$

There are only two factorisations of 51, 17×3 and 51×1 . Equating the larger factor with $u + v$ and the smaller with $u - v$, the first factorisation gives $u = 10, v = 7$. But then $y = 0$. The second factorisation gives $u = 26, v = 25$. For these we get $y = \pm\sqrt{625 - 49} = \pm\sqrt{476} = \pm 24$. So there are two points P on the y -axis, viz. $(0, 24)$ and $(0, -24)$ for which the distances PA, PO, PB are all integers. The distances are, respectively, 26, 24 and 25. ■

A simple problem requiring little more than the Pythagoras theorem and factorisation of integers. It is not clear what is gained by stipulating that A and B lie on the opposite sides of the y -axis.

Q.29 Let $G(x) = \int_{-x^3}^{x^3} f(t) dt$, where x is any real number and f is a continuous function such that $f(t) > 1$ for all real t . Then,

- (A) $G'(0) = 0$ and G has a local maximum or minimum at $x = 0$.
 (B) For any real number c , the equation $G(x) = c$ has a unique solution.
 (C) There exists a real number c such that $G(x) = c$ has no solution.
 (D) There exists a real number c such that $G(x) = c$ has more than one solution.

Answer and Comments: (B). Nothing is given about the function $f(t)$ other than that it is continuous and always bigger than 1. When the data is not specific but the correct answer is to be independent of the variety permissible in the data, it is often a good idea to take a specific, simple case of the data and see if at least some of the options are ruled out in it.

So, in the present problem, let us take $f(t)$ to be a constant function, say $f(t) = 2$ for all $t \in \mathbb{R}$. It satisfies all the conditions in the problem. For this f , the resulting function $G(x)$ is

$$\begin{aligned} G(x) &= \int_{-x^3}^{x^3} f(t) dt \\ &= \int_{-x^3}^{x^3} 2 dt \\ &= 4x^3 \end{aligned} \tag{1}$$

for all $x \in \mathbb{R}$. If we drop the positive constant 4, then this is a very standard example of a function which is strictly increasing, a bijection from $\mathbb{R}$ to $\mathbb{R}$ and has a critical point at 0 which is neither a maximum nor a minimum. So all options except (B) fail in this case. Hence (B) is correct.

An honest answer is not very different either. Note that $G(x)$ is an odd function of x , as follows from

$$G(-x) = \int_{x^3}^{-x^3} f(t) dt = - \int_{-x^3}^{x^3} f(t) dt = -G(x) \tag{2}$$

for all $x \in \mathbb{R}$. We now claim that $G(x)$ is strictly increasing on $[0, \infty)$. For this we could calculate $G'(x)$ by applying a generalisation of the fundamental theorem of calculus for differentiation of a function defined by an integral. (See (19) in Comment No. 12 of Chapter 17.) But that is hardly necessary. For suppose $0 \leq x < y$. Then $-y < -x$ and so

$$-y^3 < -x^3 \leq x^3 < y^3 \tag{3}$$

In other words, the interval $[-y^3, y^3]$ properly contains the interval $[-x^3, x^3]$. As the integrand $f(t)$ is positive throughout (in fact greater

than 1) we have

$$\begin{aligned}
 G(y) &= \int_{-y^3}^{y^3} f(t)dt \\
 &= \int_{-y^3}^{-x^3} f(t)dt + \int_{-x^3}^{x^3} f(t)dt + \int_{x^3}^{y^3} f(t)dt \\
 &> \int_{-x^3}^{x^3} f(t)dt = G(x)
 \end{aligned} \tag{4}$$

since the dropped integrals are both positive.

Thus we have proved that $G(x)$ is a strictly increasing, odd function of x . If we can prove that $G(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $G(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, then $G(x)$ would be a bijection from $\mathbb{R}$ to $\mathbb{R}$ and (B) would follow. Here for the first time we use that $f(t) > 1$ for all t . For then, for all $x > 0$

$$\begin{aligned}
 G(x) &= \int_{-x^3}^{x^3} f(t)dt \\
 &> \int_{-x^3}^{x^3} 1dt = 2x^3
 \end{aligned} \tag{5}$$

As $x^3 \rightarrow \infty$ as $x \rightarrow \infty$, it follows that $G(x) \rightarrow \infty$ as $x \rightarrow \infty$. Since $G(x)$ is an odd function, it also follows that $G(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.

Thus we have proved that (B) holds. ■

The question is poorly designed. Continuity of $f(t)$ is not needed for the correct option (B). Only its integrability is needed. Perhaps continuity is given to make (A) tempting. Also it is not clear what is gained by stipulating that $f(t) > 1$ for all t . Any positive constant lower bound on $f(t)$ would have worked equally well. The purpose of the question itself not clear.

Q.30 There are $2n + 1$ real numbers having the property that the sum of any n of them is less than the sum of the remaining $n + 1$. Then,

- (A) all the numbers must be positive
- (B) all the numbers must be negative
- (C) all the numbers must be equal
- (D) such a system of real numbers cannot exist

Answer and Comments: (A). As in the last question, when the data allows freedom of choice, we can sometimes exercise it to quickly dispose of some of the options. The problem does not specify the value of n . We take it as 1. Then we have three numbers, say a, b, c , with the property that the sum of any two of them exceeds the third. We are immediately reminded of the sides of a triangle where this holds. So the options (B), (C) and (D) are ruled out.

Again, we should give an honest argument which will show that (A) is *always* valid. Label the $2n + 1$ numbers as $x_0, x_1, x_2, \dots, x_{2n-1}, x_{2n}$ in an ascending order, that is

$$x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n \leq x_{n+1} \leq x_{n+2} \leq \dots \leq x_{2n} \quad (1)$$

Proving that all of these numbers are positive is equivalent to proving that $x_0 > 0$. We do this by contradiction. Suppose $x_0 \leq 0$. Since every number from x_1 to x_n is less than or equal to every number from x_{n+1} to x_{2n} , we surely have

$$x_1 + x_2 + \dots + x_n \leq x_{n+1} + x_{n+2} + \dots + x_{2n} \quad (2)$$

But, on the other hand, since $x_0 \leq 0$, we also have

$$x_0 + x_1 + x_2 + \dots + x_n \leq x_1 + x_2 + \dots + x_n \quad (3)$$

From (2) and (3), the sum of the n numbers from x_{n+1} to x_{2n} is bigger than or equal to the sum of the $n + 1$ numbers from x_0 to x_n , contradicting the hypothesis. So x_0 and hence each x_i is positive. ■

A similar but considerably more difficult problem was asked in the Putnam competition many years ago. Suppose we are given a collection of $2n + 1$ stones with the property that if we remove any one of them, the remaining $2n$ stones can be made into two groups of equal total weights each containing n stones. The problem was to show that this is possible if and only if all the stones are of equal weights. Sufficiency of the condition is, of course, trivial. The real task is to prove the necessity.

ISI BStat-BMath-UGB-2015 Paper with Comments

Q. 1 Let $0 < a_1 < a_2 < \dots < a_n$ be real numbers. Show that the equation

$$\frac{a_1}{a_1 - x} + \frac{a_1}{a_2 - x} + \dots + \frac{a_1}{a_n - x} = 2015$$

has exactly n real roots.

Solution Multiplying both the sides of the given equation by the product $(a_1 - x)(a_2 - x) \dots (a_n - x)$, we can reduce the problem to showing that a certain real polynomial of degree n has n real roots. While it is true that every polynomial of degree n has n real roots, they do not all have to be real. Moreover, unlike polynomials of degree 2, there is no easy criterion to decide if all the roots of an n -th degree polynomial are real. So this approach will not work.

For a new approach, call the L.H.S. of the given equation as $f(x)$. We study the behaviour of the function using calculus. Note first that if $x < a_1$, then the denominators of all n terms and hence the terms themselves are all positive. So we have

$$f(x) > 0 \tag{1}$$

for all $x < a_1$. Similarly, we have

$$f(x) < 0 \tag{2}$$

for all $x > a_n$, because all denominators are negative. Clearly, $f(x)$ is differentiable everywhere except at the points $a_1, a_2, \dots, a_n$ (at which it is undefined). For all $x \neq a_1, a_2, \dots, a_n$, we have

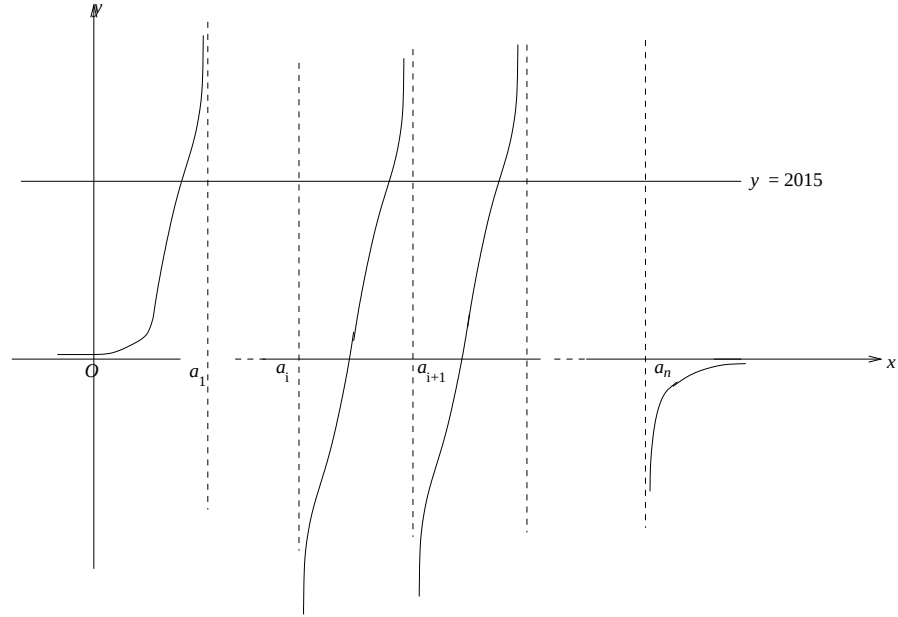
$$f'(x) = \frac{a_1}{(a_1 - x)^2} + \frac{a_2}{(a_2 - x)^2} + \dots + \frac{a_n}{(a_n - x)^2} \tag{3}$$

As all terms are positive, $f'(x) > 0$ for all $x \neq a_1, a_2, \dots, a_n$. Hence $f(x)$ is strictly increasing. This may seem to contradict (1) and (2). But it is not so. As already noted, $f(x)$ is undefined at $a_1, a_2, \dots, a_n$. In other words, f is defined on the union of the $n + 1$ disjoint intervals, $(-\infty, a_1), (a_1, a_2), (a_2, a_3), \dots, (a_{n-1}, a_n)$ and finally, (a_n, ∞) . It is true that f is strictly increasing on each of these intervals. But that does

not mean that it is increasing on their union. (This was also the major idea in Q.23 in Paper 1.)

The situation is similar to the behaviour of the function $g(x) = \tan x$. Its derivative $\sec^2 x$ is positive everywhere and so it is strictly increasing in any interval which does not contain any odd multiple of $\frac{\pi}{2}$. There is no contradiction here because as x passes through such multiples, $f(x)$ suddenly changes from very large to very small values or vice versa.

This analogy helps in sketching the graph of $f(x)$. For each i , as $x \rightarrow a_1$ from the left, $f(x) \rightarrow \infty$ because the dominant term in the sum is $\frac{a_i}{a_i - x}$ and all other terms tend to finite limits. But as $x \rightarrow a_1$ from the right, $f(x) \rightarrow -\infty$. Moreover, $f(x) \rightarrow 0$ as $x \rightarrow \pm\infty$. So the graph of $f(x)$ is (qualitatively) as shown below.



As $f(x)$ is strictly increasing in each of the $n+1$ intervals listed above, it is a bijection from (a_i, a_{i+1}) onto $\mathbb{R}$, for every $i = 1, 2, \dots, n-1$. So each of these $n-1$ interval contains exactly one root of the equation given. In addition, f maps $(-\infty, a_1)$ bijectively onto $(0, \infty)$. As 2015 is positive, this interval also contains one root. But the last interval,

viz. (a_n, ∞) , is mapped onto $(-\infty, 0)$ and contains no solution. In all the given equation has n real roots. ■

The particular number 2015 has no role whatsoever in this problem. It could as well be replaced by any real number α . (If $\alpha = 0$, there will be $n - 1$ solutions. Also if $\alpha < 0$, there will be no solution in $(-\infty, a_1)$, but there will be one in (a_n, ∞) .)

This is a good problem which will make a candidate think. Once he sees the analogy with the $\tan x$ function, the proof is easy. The essential idea in this problem and Q.23 is the same, viz. even if $f'(x) > 0$ for all x in each of several mutually disjoint open intervals, $f(x)$ need not be strictly increasing on their union. The only difference is that while in the last problem, the positivity of the derivative was given, in the present problem it comes only after some work. So Q.23 was like a forerunner to the present problem. Whether this was intentional or accidental is anybody's guess.

Q. 2 Let $\mathbb{R}$ denote the set of real numbers. Find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ satisfying

$$|f(x) - f(y)| = 2|x - y|$$

for all $x, y \in \mathbb{R}$. Justify your answer.

Solution: We first claim that f is one-to-one. This is easy because for any two x, y , $f(x) = f(y)$ implies $|f(x) - f(y)| = 0$ and hence $2|x - y| = 0$ from the data. But that further means that $|x - y| = 0$ and hence $x = y$.

Next, we claim that f is continuous everywhere. This is also easy. Indeed, for every fixed x ,

$$|f(x + h) - f(x)| = 2|h| \tag{1}$$

As $h \rightarrow 0$, $|h| \rightarrow 0$ and so $f(x + h) - f(x) \rightarrow 0$. (For an $\epsilon - \delta$ proof, given any $\epsilon > 0$, take $\delta = \frac{\epsilon}{2}$. Then (1) shows that $|f(x + h) - f(x)| < \epsilon$ whenever $|h| < \delta$. Actually, this is a rare case, where the *delta* which works for a given $\epsilon > 0$ in the definition of a limit can be determined so explicitly in terms of ϵ . Note more importantly, that the δ depends only on ϵ and not on the point x . For this reason, $f(x)$ is an example

of what is called a **uniformly continuous** function. But we shall not go into it.)

By the Intermediate Value Property, the function $f : \mathbb{R} \rightarrow \mathbb{R}$, being one-to-one and continuous, is either strictly increasing or strictly decreasing on the entire $\mathbb{R}$. (For a proof see Comment No. 6 of Chapter 16.) So, we have to make two cases.

First suppose that $f(x)$ is strictly increasing. Then for every $h > 0$ we have

$$f(x+h) - f(x) = |f(x+h) - f(x)| = 2|h| = 2h \quad (2)$$

Dividing by h and taking limit as $h \rightarrow 0^+$, we get that the right handed derivative $f'_+(x)$ exists and equals 2 at every x .

If $h < 0$, then also (2) holds, because

$$f(x+h) - f(x) = -|f(x+h) - f(x)| = -2|h| = 2h \quad (3)$$

So, the left handed derivative $f'_-(x)$ also exists and equals 2. Put together we get that $f(x)$ is differentiable everywhere and

$$f'(x) = 2 \quad (4)$$

for all $x \in \mathbb{R}$. Integrating both the sides (or applying Lagrange's MVT)

$$f(x) - f(0) = 2x \quad (5)$$

for all $x \in \mathbb{R}$. We do not know what $f(0)$ is. It is some constant, say c .

So, when $f(x)$ is strictly increasing, it is of the form

$$f(x) = 2x + c \quad (6)$$

for some constant c .

A similar argument can be given for the other case, viz. where $f(x)$ is strictly decreasing on $\mathbb{R}$. But that is hardly necessary, because if $f(x)$ is increasing, then $-f$ is increasing and satisfies the given condition for all x, y . Therefore

$$-f(x) = 2x + d \quad (7)$$

for some constant d . Put differently,

$$f(x) = -2x - d \quad (8)$$

We might as well denote $-d$ by c , because the two possibilities can never hold together.

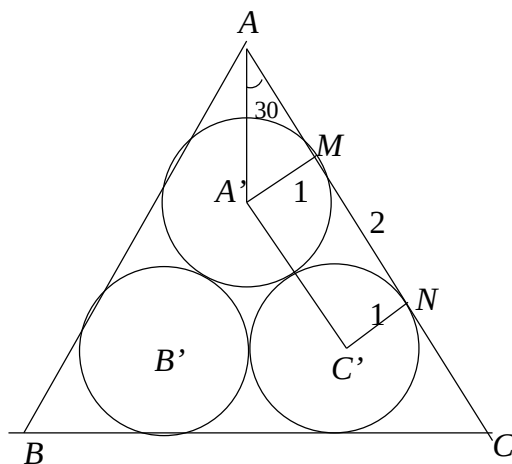
Summing up, all functions which satisfy the given condition are of the form $f(x) = \pm 2x + c$ for some constant c . ■

The converse is, of course, trivial. That is if $f(x)$ is of the form $\pm 2x + c$, then it satisfies the given condition trivially. But proving that these are the only functions that do so is non-trivial.

The condition in the statement of the problem is an example of what is called a **functional equation**. See Chapter 20 for more on them.

- Q. 3** Three circles of unit radius tangentially touch each other in the plane. Consider the triangle enclosing them such that each side of the triangle is tangential to two of these three circles. Find the length of each side of the triangle.

Solution: The question paper contains a diagram which we reproduce below along with our own notations and some constructions.



Let the triangle formed by the tangents be ABC . Let A', B', C' be the centres of the three circles. Then the sides of $\triangle A'B'C'$ are parallel

to those of $\triangle ABC$ since all the three circles are of equal radii. Drop perpendiculars $A'M$ and $C'N$ from A', C' to the side AC . Join AA' . By symmetry, AA' divides $\angle A$. So

$$AM = A'M \tan(A/2) = 1 \times \tan 30^\circ \quad (1)$$

Similarly,

$$NC = \tan 30^\circ \quad (2)$$

Further $A'C'NM$ is a rectangle with $A'C' = 2$, being the sum of the radii of two circles touching each other externally. So,

$$MN = A'C' = 2 \quad (3)$$

Adding (1), (2) and (3),

$$AC = 2 + 2 \tan 30^\circ = 2 + \frac{2}{\sqrt{3}} \quad (4)$$

By symmetry, this is also the length of each of the other two sides. ■

More generally, the three mutually touching circles could be of different radii, r_1, r_2, r_3 . Then the sides of the triangle can be expressed in terms of r_1, r_2, r_3 . But the expressions would be fairly complicated and the work needed would require some identities about triangles. (See the Main Problem of Chapter 11, which, however, asks for the inradius of the triangle $A'B'C'$ in terms of r_1, r_2, r_3 .) The paper setters have realised this and made the problem simple. It is a very good geometric problem which unfolds itself once you come up with the right construction.

- Q. 4** Let a and b be real numbers. Define a function $f : \mathbb{R} \rightarrow \mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers, by the formula $f(x) = x^2 + ax + b$. Assume that the graph of f intersects the co-ordinate axes in three distinct points. Prove that the circle passing through these three points also passes through the point $(0, 1)$.

Solution: The problem is more on geometry than on quadratic equations. For *any* function $f(x)$, the only point where its graph cuts the y -axis is $(0, f(0))$ provided $f(0)$ is defined. Call this point B . In the

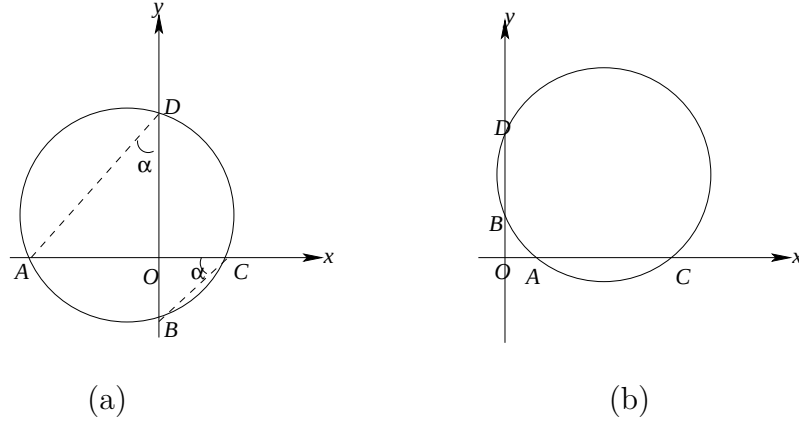
present problem, $B = (0, b)$. The other two points of intersection with the axes must lie on the x -axis and hence must represent the roots, say α and β of the quadratic $x^2 + ax + b$. Denote the points $(\alpha, 0)$ and $(\beta, 0)$ by A and C . Then we have

$$\alpha\beta = b \quad (1)$$

If we ignore the signs of α and β , then geometrically this means

$$OA \cdot OC = OB \quad (2)$$

Now, suppose that the circle cuts the y -axis at the point D again. Then by a classic result about two chords AC and BD intersecting at a point O , we also have



$$OA \cdot OC = OB \cdot OD \quad (3)$$

(2) and (3) together imply

$$OD = 1 \quad (4)$$

Since we are ignoring signs, this does not necessarily imply that D is the point $(0, 1)$. All we can say is that $D = (0, \pm 1)$. To make the sign distinction, we need to look at the sign of b . If $b < 0$, then the point B lies below the x -axis as in (a). But then by (1), α and β are of opposite signs. So A and C lie on opposite sides of the x -axis. Which one lies

on which side is immaterial because in either case the chords AC and BD intersect at the point O inside the circle. But then B and D also lie on opposite sides of the point of intersection O . So $D = (0, 1)$.

On the other hand if $b > 0$, then B lies above the x -axis and A, C lie on the same side of the y -axis as in (b). Regardless of which side they lie, the point O of intersection now lies outside the circle. Therefore the points B and D lie on the same side of the x -axis. So in this case too, $D = (0, 1)$. ■

The geometric result we quoted here is a consequence of the cyclicity of the quadrilateral $ABCD$, which makes the angles $\angle ADB$ and $\angle ACB$ (shown by α in (a)) equal and hence the triangles OAD and OCB similar. It is immaterial whether the chords intersect at right angles as in the present problem. It is also valid when the chords meet at a point outside the circle as in (b). But usually in elementary geometry books, separate proofs are given for the two cases.

The problem can also be tackled purely algebraically. In fact, the algebraic approach has the advantage that it works regardless of whether b is positive or negative and hence whether A and C are of the same or the opposite signs. But such proofs lack the visual appeal of geometry. You never ‘see’ what is going on. You merely manipulate some equations and the answer pops out.

We begin with the general equation of a circle, viz.

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (5)$$

In our present problem, we are given that the circle passes through the points $(\alpha, 0)$, $(\beta, 0)$ and $(0, b)$. This gives us three equations, viz.

$$\alpha^2 + 2g\alpha + c = 0 \quad (6)$$

$$\beta^2 + 2g\beta + c = 0 \quad (7)$$

$$\text{and } b^2 + 2fb + c = 0 \quad (8)$$

This is a system of three equations in the three unknowns g, f and c and if we solve it, we shall know the circle completely. But that is really not needed. Our goal is to show that the point $(0, 1)$ lies on this circle, i.e.

$$1 + 2f + c = 0 \quad (9)$$

and we can get this from the three equations without actually solving them. We first eliminate g between (6) and (7) by multiplying the former by β , the latter by α and then subtracting. This gives

$$\alpha\beta(\alpha - \beta) = c(\alpha - \beta) \quad (10)$$

As the roots are distinct, we cancel $\alpha - \beta$ and use (1) to get

$$b = c \quad (11)$$

Putting this into (8)

$$c^2 + 2fc + c = 0 \quad (12)$$

Clearly $c \neq 0$ as otherwise $b = 0$ too and the three points A, B, C would be collinear. So cancelling c , we get

$$c + 2f + 1 = 0 \quad (13)$$

which is exactly (9).

Since we have a purely algebraic solution without requiring the theorem from pure geometry about intersecting chords of a circle, it is natural to inquire if that theorem itself can be proved algebraically. The answer is ‘yes’ and is easy if the chords intersect each other at right angles, because we can take the chords to lie along the coordinate axes. Then their point of intersection is the origin. Setting $y = 0$ in (5), the two ends of one of the chords will be of the form $(x_1, 0)$ and $(x_2, 0)$ where x_1, x_2 the roots of the quadratic $x^2 + 2gx + c = 0$. Similarly, those of the other chord will be $(0, y_1)$ and $(0, y_2)$ where y_1, y_2 are the roots of the quadratic $y^2 + 2fy + c = 0$. The result follows because the product of the two roots is the same, viz. c for both the quadratics.

But when the chords meet at some other angle, the calculations get more complicated. So the pure geometry result is superior in that the proof is equally easy regardless of the angle of intersection of the two chords. The same is the case with the pure geometry result that the angle subtended by a chord at the centre is twice the angle subtended at any point on the opposite arc. This is not a hard and fast rule. Sometimes algebraic proofs are easier, for example, the proof of the concurrency of the three altitudes of a triangle. Moreover, figures are

generally not necessary for algebraic proofs. In friendly debates, the lovers of geometry scoff at the algebraic methods as inelegant donkey work while those of algebra chide the geometric approach as smart hand waving passed off as elegance. The true lovers of mathematics master both and use whichever is more suited for a particular problem.

- Q. 5** Find all positive integers n for which $5^n + 1$ is divisible by 7. Justify your answer.

Answer and Solution: All integers of the form $3 + 6k$, where $k = 0, 1, 2, \dots$

The best thing is to begin by trial. For $n = 1, 2, 3$ the integers $5^n + 1$ are, 6, 26 and 126 respectively. The first two are not divisible by 7 but the third one is since $126 = 18 \times 7$. So $n = 3$ is a correct answer. But how do we find the others? And moreover, how do we prove that they are the only ones?

Before answering these, let us carry out the experimentation a little further. Consider $5^4 + 1$ which equals 626. We can see directly that it is not divisible by 7. But there is a short cut. We rewrite it as $500 + 126$. Since the second term is divisible by 7 but the first one is not, the sum is not divisible. (Some people fumble at swallowing this. The proof is in one line. Write 500 as $626 - 126$. If 626 and 126 were both divisible by 7, so would be 500 which is not the case.)

The next in line $5^5 + 1 = 3126$. By the same argument as before, we see that 3126 is not divisible by 7. The next one, $5^6 + 1$ equals 15626. We rewrite this as $15500 + 126$ and see that it is not divisible by 7.

But as the exponent n gets bigger and bigger the numbers $5^n + 1$ will increase and testing their divisibility by 7 will get more difficult. And, at any rate, we cannot do this for infinitely many n . So we look for a short cut. So let us see if we can prove the non-divisibility of 626, 3126 and 15626 by some other means. We have already proved that $5^3 + 1$ is divisible by 7. So we can write

$$5^3 = 7m - 1 \tag{1}$$

for some integer m . (The exact value of m is 18 but that is not so important. What matters more is that it is an integer.) As a result,

we can write

$$5^4 + 1 = 5(5^3) + 1 = 5(7m - 1) + 1 = 35m - 4 \quad (2)$$

Since 35 is divisible by 7 but -4 is not, we see very effortlessly that $5^4 + 1$ is not divisible by 7. The reasoning for the next two numbers is similar.

$$\begin{aligned} 5^5 + 1 &= 25(5^3) + 1 = 25(7m - 1) + 1 = 25 \times 7m + 24 \quad (3) \\ \text{and } 5^6 + 1 &= (5^3)^2 + 1 = (7m - 1)^2 + 1 = 49m^2 - 14m + 2 \quad (4) \end{aligned}$$

The next number $5^7 + 1$, can be tackled using (4).

$$5^7 + 1 = 5(5^6) + 1 = 5(7m - 1)^2 + 5 = 5(49m^2 - 14m) + 6 \quad (5)$$

So again it is not divisible by 7.

Let us recast our work so far in terms of the remainder which the successive powers of 5 leave when divided by 7. Starting from 5^1 , these remainders are

$$5, 4, 6, 2, 3, 1, 5 \quad (6)$$

(To see the fourth entry, we rewrite (2) as $5^4 = 35m - 5 = 7(5m - 1) + 2$.)

The problem is equivalent to asking for which integers n , 5^n leaves a remainder 6. From the list above, we see that among the first seven powers, 5^3 is the only such power.

The key to the solution is the observation, that each remainder determines the next one very easily. For example, take the fourth remainder 2. Then $5^4 = 7q + 2$ for some integer q . But then $5^5 = 35q + 10 = 7(5q + 1) + 3$. We can actually drop q and instead of multiplying $7q + 2$ by 5, we simply multiply the remainder 2 by 5 to get 10 which leaves a remainder 3. Similarly, $3 \times 5 = 15$ leaves a remainder 1. That is why the term following 3 is 1.

Since the first and the seventh remainders are the same (viz. 5 each), the eighth remainder will be the same as the second one, viz. 4. So the remainders left by 5^n when divided by 7 will repeat in a cycle of length 6. As a result, the integers n for which 5^n leaves a remainder 6 (and hence $5^n + 1$ is divisible by 7) are integers of the form $3 + 6k$ for some integer $k = 0, 1, 2, 3, \dots$ ■

A very good problem which tests a candidate's ability to experiment, observe some pattern, guess the correct answer and, finally, prove that the guess is correct. The more formal term for a remainder is a 'residue'. The entries in the sequence (6) are what are called the **residue classes** of the successive powers of 5, modulo 7. We have avoided this terminology and instead given the argument in simpler terms. But chances are that some candidates already know about residue arithmetic. For them the problem will be a lot easier.

- Q. 6** Let $p(x) = x^7 + x^6 + b_5x^5 + \dots + b_1x + b_0$ and $q(x) = x^5 + c_4x^4 + \dots + c_1x + c_0$ be polynomials with integer coefficients. Assume that $p(i) = q(i)$ for integers $i = 1, 2, \dots, 6$. Then, show that there exists a negative integer r such that $p(r) = q(r)$.

Solution: The hypothesis as well as the conclusion deals with the equality of $p(x)$ and $q(x)$ for certain integer values of x . This can be conveniently transformed by considering their difference defined by

$$\begin{aligned} f(x) &= p(x) - q(x) \\ &= x^7 + x^6 + (b_5 - 1)x^5 + (b_4 - c_4)x^4 + \dots + (b_1 - c_1)x + (b_0 - c_0) \end{aligned}$$

Then we are given that $f(i) = 0$ for $i = 1, 2, \dots, 6$. In other words, $1, 2, \dots, 6$ are among the roots of $f(x)$. Being a polynomial of degree 7, $f(x)$ has seven roots. Let the missing root be r . By a well-known formula, the sum of the roots of a polynomial equals the negative of the ratio of the coefficient of the second highest power (in the present case, x^6) by the leading coefficient, which is 1 here. So

$$r + 1 + 2 + 3 + 4 + 5 + 6 = -\frac{1}{1} = -1 \quad (7)$$

Hence $r = -1 - (1 + 2 + 3 + 4 + 5 + 6) = -22$ is the missing root of $f(x)$ and hence a common root of $p(x)$ and $q(x)$. Clearly r is a negative integer. ■

A disappointing problem, in which nothing is left once the idea of taking the difference of the two polynomials strikes. The computational work needed is less than that for many questions in Paper 1. Still, it is difficult to ask this question as an MCQ, because then -22 will have to be one of the options and that itself will give a hint to the solution.

Q. 7 Let $S = \{1, 2, \dots, j\}$. For every non-empty subset A of S , let $m(A)$ denote the maximum element of A . Then, show that

$$\sum m(A) = (j-1)2^j + 1$$

where the summation in the left hand side of the above equation is taken over all non-empty subsets A of S .

Solution: Denote the sum on the L.H.S. by S_j . It is generally a good idea in such problems to work out a few simple cases by hand. Sometimes they give a clue to the general case. The case $j = 1$ is trivial since $\{1\}$ is the only non-empty subset here and its maximum element is 1 itself. Take $S = \{1, 2\}$. Then the number of non-empty subsets is $2^2 - 1 = 3$. If we write their elements in ascending order, they are

$$\{1\}, \{2\}, \{1, 2\} \quad (1)$$

Hence

$$S_2 = 1 + 2 + 2 = 5 \quad (2)$$

and we see that this indeed equals $(2-1)2^2 + 1$. The next case is $j = 3$. This time there are seven non-empty subsets:

$$\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\} \text{ and } \{1, 2, 3\} \quad (3)$$

So again, by a direct count,

$$S_3 = 1 + 2 + 3 + 2 + 3 + 3 + 3 = 17 = (3-1)2^3 + 1 \quad (4)$$

We have not proved the assertion for the general case. But we are now convinced that it is true. To see how to go further, let us obtain (4) from (2). We change the order of listing the subsets in (3), so that subsets not containing the element 3 appear first. So, the seven non-empty subsets of $\{1, 2, 3\}$ are

$$\{1\}, \{2\}, \{1, 2\}, \{3\}, \{1, 3\}, \{2, 3\} \text{ and } \{1, 2, 3\} \quad (5)$$

Then (2) already gives us the sum of the maximum elements of the first three subsets. All the remaining four subsets have 3 as their maximum elements. So they contribute 12. Hence

$$S_3 = S_2 + 2^2 \cdot 3 \quad (6)$$

It is obvious that a similar equality will hold if we replace 2 by any positive integer. That is, we shall have

$$S_{j+1} = S_j + 2^j(j+1) \quad (7)$$

The proof is also the same. We classify all non-empty subsets of the set $\{1, 2, \dots, j, j+1\}$ into two types, Those that don't contain $j+1$ and those that do. The former can be thought of as non-empty subsets of the set $\{1, 2, \dots, j\}$ and so the sum of their maximum elements is S_j . Of the latter type, there are 2^j subsets. They are obtained by adding the element $j+1$ to all subsets of $\{1, 2, \dots, j\}$. (Note that the empty subset is also included, by adding $j+1$ to it we get the non-empty singleton subset $\{j+1\}$).

Because of (7), we can prove the assertion by induction on j . The case $j = 1$ is trivially true as already noted. Now suppose that the result holds for $j = k$. To prove it for $j = k+1$, we apply (7) with $j = k$ and the induction hypothesis to get

$$\begin{aligned} S_{k+1} &= S_k + 2^k(k+1) \\ &= (k-1)2^k + 1 + 2^k(k+1) \\ &= (k-1+k+1)2^k + 1 \\ &= 2k2^k + 1 \\ &= (k+1-1)2^{k+1} + 1 \end{aligned} \quad (8)$$

which completes the induction. ■

Another simple problem. But better than the last one because a candidate's ability to experiment, recognise a pattern, make a guess and prove it is tested.

- Q.8** (a) Let $m_1 < m_2 < \dots < m_k$ be positive integers such that $\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_k}$ are in arithmetic progression. Then prove that $k < m_1 + 2$.
- (b) For any integer $k > 0$, give an example of a sequence of k positive integers whose reciprocals are in arithmetic progression.

Solution: We are given that the reciprocals $\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_k}$ are in an arithmetic progression. This is a descending progression because the

common difference $\frac{1}{m_2} - \frac{1}{m_1}$ is negative (as we are given $m_1 < m_2$), But all the terms are positive. Such an A.P. cannot go on forever unlike a descending A.P. whose terms belong to a set which is not bounded below (e.g. the set of all integers, all reals, all negative numbers). This is intuitively obvious because between a positive real number x and 0, you can only squeeze in a finite number of evenly spaced points at a given distance apart. More precisely, if d is the common difference and x is the first term, then $x + d(k-1) > 0$ puts an upper bound $1 - \frac{x}{d}$ on k . In the present problem. $x = \frac{1}{m_1}$ and $d = \frac{1}{m_2} - \frac{1}{m_1}$. Unfortunately, the upper bound on k we get from this is only $1 + \frac{m_2}{m_2 - m_1}$ which is not as strong as that demanded by (a). Moreover it involves both m_1 and m_2 whereas the question asks us to prove an upper bound which is solely in terms of m_1 .

The failure of this simplistic approach is predictable because it makes little use of the fact that the terms of the progression are reciprocals of positive integers. It is applicable for a descending A.P. whose terms are any positive real numbers. It stands to common sense that a more general result when applied to a particular case cannot be as deep as one which is possible in that special case by more specific arguments. (A refinement of this approach which works will be given at the end.)

So we need to tackle the problem more delicately. Arithmetic progressions of integers are far more common and easier to handle than those whose elements are reciprocals of positive integers. So let us see if the given A.P.. viz., $\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_k}$ can be converted to an A.P. of integers. It turns out that this can be done in a very natural way.

The basic idea is that an arithmetic progression remains an A.P. if all terms are multiplied by some constant. So, suppose that $\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_k}$ are in A.P. Let M be the l.c.m. (= least common multiple) of $m_1, m_2, \dots, m_k$. Then $\frac{M}{m_1}, \frac{M}{m_2}, \dots, \frac{M}{m_k}$ are in A.P. But these are all integers, say $u_1, u_2, \dots, u_k$. Thus we get an A.P. of positive integers

$$u_1 > u_2 > u_3 > \dots > u_k \quad (1)$$

Note that the common difference d here is a negative integer.

Conversely, starting with any A.P. of positive integers with a negative common difference, we can divide each term by their l.c.m., say U

and get an A.P. whose entries are reciprocals of positive integers. For example, take the progression

$$8, 6, 4, 2 \quad (2)$$

Here the l.c.m. U is 24. Dividing by it we get the A.P.

$$\frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{12} \quad (3)$$

Here the integers m_1, m_2, m_3, m_4 are 3, 4, 6 and 12 respectively. Note that their l.c.m. is also 24 and so if we multiply all terms in (3) by 24 we get back the A.P. (2).

(Of course, we can also divide every term in (2) by any multiple of 24. But dividing by 24 gives an A.P. of reciprocals of integers which are as small as possible.)

This gives an easy answer to (b). We start with any progression of length k of positive integers and divide each term by the l.c.m. of these k integers.

Let us now turn to Part (a). Again, we may suppose that the integers $m_1, m_2, \dots, m_k$ are of the form $\frac{U}{u_1}, \frac{U}{u_2}, \dots, \frac{U}{u_k}$ where the integers $u_1, u_2, \dots, u_k$ satisfy (1) and

$$u_i - u_{i-1} = d \quad (4)$$

for $i = 2, 3, \dots, k$, the integer d being negative, and U is the l.c.m. of $u_1, u_2, \dots, u_k$. The inequality to be proved reduces to

$$k \leq m_1 + 1 = \frac{U}{u_1} + 1 = \frac{U + u_1}{u_1} \quad (5)$$

and hence further to

$$(k - 1)u_1 \leq U \quad (6)$$

Note that equality may hold as see from (2), where $k = 4, U = 24$ and $u_1 = 8$. Let us analyse why equality holds here. Surely, we expect that the l.c.m. U is bigger than u_1 , because although u_1 divides U , U may also have factors coming from other u_i 's, viz. u_2, u_3 and u_4 . But in (2),

the last two terms, viz. 4 and 2 contribute nothing new to U since they are factors of u_1 which is 8. They merely increase the length, viz. k of the progression. It is only the term u_2 , which is 6 which shoots up U . Again, it does not boost U by 6, but only by 3, because the other factor of it (viz. 2) is already absorbed in u_1 . If u_2 were relatively prime to u_1 , then U will equal at least $u_1 u_2$.

This suggests that we should normalise the progression (2). We do so by cancelling common factors of u_1 and u_2 . In (2) this common factor is 2. It is also a factor of the remaining terms of the progression. So dividing all of them by 2, we get a new A.P. instead of (2), viz.

$$4, 3, 2, 1 \quad (7)$$

Now the l.c.m. is not 24 but only 12. Still, if we divide all terms by 12, the resulting A.P. of fractions will be the same as (3). As a result, the integers m_1, m_2, m_3, m_4 remain the same, viz. 3, 4, 6 and 12.

We carry out a similar normalisation for the A.P. (1) which we took as a possible source to generate the given integers $m_1, m_2, \dots, m_k$. But this source is not unique. Any multiple of it will also generate the same integers $m_1, m_2, \dots, m_k$, because the l.c.m. will also get multiplied by the same factor and hence its ratios to the terms in (1) will remain the same.

We now go the other way. That is, we see if the terms in (1) can be reduced in some way. Note that if r is a common factor of u_1 and u_2 , then it will also be a factor of d since $d = u_2 - u_1$. But then it will also be a factor of u_3 since $u_3 = u_2 + d$, and hence further of u_4 too because $u_4 = u_3 + d$. Indeed this will go on till u_k . So we take r to be the g.c.d. (= greatest common divisor) of u_1 and u_2 and divide each u_i by r to get a new A.P.

$$v_1 > v_2 > v_3 > \dots > v_k \quad (8)$$

where $v_i = \frac{u_i}{r}$ for $i = 1, 2, \dots, k$. Let V be the l.c.m. of these $v_1, v_2, \dots, v_k$. Then $V = \frac{U}{r}$ and so the ratios $\frac{V}{v_1}, \frac{V}{v_2}, \dots, \frac{V}{v_k}$ will remain the same as $\frac{U}{u_1}, \frac{U}{u_2}, \dots, \frac{U}{u_k}$ and hence will generate the same integers $m_1, m_2, \dots, m_k$.

But there is one respect in which (8) is superior to (1). In (8), the first two terms, viz. v_1 and v_2 have no common factors, by construction, because they have been obtained by dividing u_1 and u_2 by their g.c.d. r . We can't be so sure of the relationship between the remaining pairs of terms. For example, after converting (2) to (7), the first and the third terms do have a common factor, viz. 2. Indeed, any A.P. (with integer terms) of length 4 or more is bound to contain at least two even terms. But it is good enough for our purpose. The inequality (6) which we are after now reduces to

$$(k-1)v_1 \leq V \quad (9)$$

As before, each v_i divides V . But the advantage this time is that since v_2 has no common factors with v_1 , not only v_1, v_2 each divide V , but their product v_1v_2 also divides V . (This follows by considering the prime power factorisations of v_1 and v_2 , which cannot have any common primes.) In particular, $V \geq v_1v_2$. Therefore we shall be through if we can prove that

$$k-1 \leq v_2 \quad (10)$$

And this is true for an unexpectedly simple reason. (8) is a strictly descending sequence of positive integers. The last term is at least 1. Hence the second term v_2 is at least $k-1$, because there are at least $k-2$ positive integers all less than it!

This completes the proof of (a). As (b) was already proved, the solution is complete. As remarked at the beginning, if we merely wanted to prove (a), there is a way out which makes a crucial use of the fact that $m_1, m_2, \dots, m_k$ are positive integers. As before, we let d be the common difference of the A.P. $\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_k}$ where $m_1, m_2, \dots, m_k$ are positive integers in an ascending order. Then,

$$d = \frac{1}{m_2} - \frac{1}{m_1} \quad (11)$$

We now use the fact that since m_1 and m_2 are integers and $m_2 > m_1$, we must have $m_2 \geq m_1 + 1$. Therefore $\frac{1}{m_2} \leq \frac{1}{m_1+1}$. Putting this into (11), we get

$$d \leq \frac{1}{m_1+1} - \frac{1}{m_1} = \frac{-1}{m_1(m_1+1)} \quad (12)$$

Note that $d < 0$ as the reciprocals are in a descending order. Now, the last term of the A.P., viz. $\frac{1}{m_k}$ is also positive and hence

$$\frac{1}{m_k} = \frac{1}{m_1} + (k-1)d > 0 \quad (13)$$

Combining this with (12) gives

$$\frac{1}{m_1} - \frac{k-1}{m_1(m_1+1)} \geq \frac{1}{m_1} + (k-1)d > 0 \quad (14)$$

Hence

$$\frac{1}{m_1} > \frac{k-1}{m_1(m_1+1)} \quad (15)$$

and therefore

$$m_1 + 1 > k - 1 \quad (16)$$

i.e. $k < m_1 + 2$ as was to be proved. ■

This argument, whose essence is that the difference of two (unequal) integers is at least 1, is a lot more direct than the earlier argument where we converted the A.P. $\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_k}$ to an A.P. whose entries are integers by multiplying it throughout by some positive integer. But it does not readily give (b). In our earlier approach, (b) followed very easily but (a) required some number theory (specifically, the fact that if v_1, v_2 are relatively prime divisors of V then $v_1 v_2$ also divides V).

This problem is, by far, the best problem, not only in Paper 2 but in both the papers put together. It tests the ability to recognize why a simplistic approach will not work, to look for a suitable transformation of the problem and solve this transformed problem by working out a few simple cases to get an idea of the general method. In the present problem, the essential idea behind this transformation is very simple, viz. that an A.P. remains an A.P. when all terms are multiplied or divided by a common factor. And the climax comes because of the unexpectedly simple proof of the inequality (10).

CONCLUDING REMARKS

Although my commentaries on JEE are written chronologically, that is not the case with the commentaries on ISI B.Sc. While writing the commentary on JEE 2017, my attention was drawn to the ISI papers of 2016. (Those of 2017 were not officially released then.) When I compared JEE 2017 with ISI 2016. I found that despite the limited scope of the syllabus, the latter contained many imaginatively designed problems. So I decided to write a commentary on ISI 2016 papers. This was an enjoyable experience. This was followed by ISI 2014, which too, was equally enjoyable. (There was a shade of disillusionment when some reader pointed out that Q.27 of 2014 Paper 1 about minimisation of $|\sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x|$) was a straight replica of a 2003 Putnam problem. But this revelation came only after the present ISI 2015 commentary was completed.)

As a result, expectations were high when I set out to write the commentary on ISI 2015. But Paper 1 was a big disappointment. In fact, some questions (e.g. Q.2 about the number of one-to-one functions from one set to another, Q.3 about the maximum area of a triangle with two given sides, Q.5 about an obvious application of the A.M.-G.M. inequality, Q.14 about the number of integral points on a line segment in a plane, Q.15 about the trigonometric ratios of the angles in a triangle, Q.21 about finding triplets of primes adding to 30, Q.22 about the integral of an odd function, Q.24 about complex representation of the vertices of a regular polygon) were so hopelessly below the expectations that I thought of quitting.

Another irritating feature of Paper 1 was not only the duplication, but the triplication of the A.M.-G.M. inequality. Q.5, Q.26 and Q.27 are all based on it.

A welcome feature of Paper 1 is that two problems (Q.9 about linear dependence of functions and Q.10 about multiplicatively closed subsets) begin by defining a new concept. The ability to correctly grasp definitions is often lost in the preparation based on a plethora of formulas. So a few questions designed to simply test this ability are welcome, even though mathematically the questions themselves may be very simple.

Q.16 about finding the number of subsets of even cardinalities is imaginatively designed. Unfortunately, it can also be done in a more straightforward manner and so the creativity of the paper setters is likely to be lost on most candidates. Q.20 (about a ten digit integer) is more like an entertaining puzzle. The trigonometry problem (Q.13) about finding the length of an inverse common tangent of two given circles is good, But, as commented there,

it would have found a better place in Paper 2 as a problem of constructing this tangent. It is high time that this lost art is revived.

Paper 2 was comparatively better than Paper 1. The first four problems are fairly good. Problem 5 about residues of powers of 5 modulo 7 is still better. But Q.6 (about common roots of two polynomials) was very disappointing. But the last problem (about reciprocals of integers in an A.P.) is by far the best problem of the entire paper. It is only because of this problem that I feel tempted to wait for the release of ISI 2017 papers and to write about them. It is a sad mystery why they are still not in the public domain.